

INSTITUT MINES-TÉLÉCOM, SIO
Second-year engineering internship

Predicting Which Antenna a Mobile Phone Will Use Next

A language model retrained on one day of network records

Olivier Recher

<i>Host institution</i>	Czech Technical University in Prague (ČVUT)
<i>Academic supervisor</i>	Robert Besták
<i>Period</i>	29 June – 28 August 2026

Abstract

A mobile phone is always attached to one antenna, and the operator writes a line down each time it changes. This report asks whether a language model, retrained on those anonymous lines, can guess which antenna a person’s phone will be attached to next, better than the rule “it is still on the same one as a moment ago”. That rule, called the *persistence baseline* here, is the yardstick for every claim in this document, and it is a hard one: on an ordinary day of records it is right 69.2 times out of a hundred, and for five of the eleven iterations of this internship nothing I trained beat it.

The material is fifteen daily exports covering the Karlovy Vary study area, 369 antennas mounted on 113 masts, with one day of records per person. On the thousand people held out of the largest set, the retrained model answers correctly **4.3 more times out of a hundred** than the one-line rule, 54.5 against 50.2. What produced that gain was neither a bigger model nor more data, but conditioning on *who the person is*. Two further results frame it: population size stops paying at around two thousand people, and 22.4 correct answers per hundred separate the stay-at-home user from the morning-moving user, against 0.4 across the entire range of training sizes. Difficulty lives in the person, not in the dataset.

Keywords: human mobility prediction, call detail records,¹ next-cell prediction, large language models, LoRA, QLoRA, behavioural clustering, scaling.

¹**CDR**, *call detail record*: the line an operator writes whenever a phone and the network interact. It holds an anonymous identifier, a time, and the antenna in use. No content of any kind.

The internship in two pages

The goal. Predict which antenna a mobile phone will be attached to next, from the beginning of that person's day and nothing else. An operator wants this because it decides where capacity is needed, and because a connection prepared before it is required is a connection that does not drop.

The data. Fifteen daily exports from a Czech operator covering the Karlovy Vary area, 12 to 26 March 2014. 369 antennas mounted on 113 masts, and one line per person per day: an anonymous number, then the antennas that person's phone used with the time of each. No names, no numbers, no content. A day file holds between 80,217 and 99,504 people.

The competitor. The rule "*the person is still on the same antenna as a moment ago*". One line of code, no training. On a day file taken as it comes it is right 69.2 times out of a hundred, because people spend most of their time not going anywhere. Everything in this report is a distance to that rule, never an absolute score.

What I built. A small general-purpose language model, retrained on those days until it stops being a writer of English and becomes a predictor of movement in one Czech region. Every run was trained on one free graphics card. The whole instrument was fixed in the first week and never changed afterwards: what changed, eleven times, was the data and the training signal.

What I actually did, week by week. Table 1 is the internship. Each row is one iteration: which people it was measured on, the one thing I changed that week, and how far the trained model landed from the one-line rule.

Table 1: The eleven iterations, on one axis

It.	People	What I changed that week	Graded on	Rule	Model	Gap
1	RAW DAY	built the pipeline end to end	last fifth	69.2 %	16.9 %	−52.3
2	RAW DAY	read the best saved copy	last fifth	69.2 %	64.4 %	−4.8
3	RAW DAY, 5,000 users	ten times more people	last fifth	81.6 %	76.0 %	−5.6
4	FILTERED DAY	dropped near-silent people	last fifth	79.0 %	80.1 %	+1.1
5	MOBILE-USER DAY	kept only people who move	last fifth	45.5 %	44.6 %	−0.9
6	MOBILE-USER DAY	gave the model the hour of day	last fifth	45.5 %	44.2 %	−1.3
7	MOBILE-USER DAY	named the kind of user in the text	whole day	43.1 %	41.3 %	−1.8
8	MOBILE-USER DAY	one set of numbers per kind of user	whole day	43.1 %	46.8 %	+3.7
9	2,000-USER DAY	1,900 training people	last fifth	48.2 %	52.4 %	+4.2
9	LEVELLED CORPUS	stretched every day to 200 records	last fifth	76.5 %	74.9 %	−1.6
10	ELEVEN-DAY SET	21,000 training people	last fifth	50.2 %	54.5 %	+4.3
11	ELEVEN-DAY SET	<i>no new training: why the curve climbs, then stops</i>				

The only column that may be read vertically is the last one. **Gap** is how many more answers out of a hundred the model got right than the one-line rule, on that week's own test people. Five weeks below it, one accidental win at iteration 4, and a real one from iteration 8. The **Rule** column is why no other column may be compared from row to row: the set of people changed four times, and with it the difficulty of the test. "Last fifth" grades the final fifth of each test day, "whole day" every position of it one at a time, which is harder and lowers the rule on the very same people from 45.5 % to 43.1 %.

Two rows read wrong on their own. Iteration 6 won 2.6 more correct answers per hundred inside the two stretches of the day it targeted and lost more than that everywhere else, so its single number hides the mechanism that iteration 8 then used properly. Iteration 9 appears twice because two independent things were tried that week. What each iteration actually did is in Appendix A.

What came out of it. Three findings, each established in the chapter named.

1. **The model beats the rule by 4.3 more correct answers per hundred**, 54.5 % against 50.2 % on a thousand people it had never seen, and by 4.2 on a second, separate group. What produced that gain was neither a bigger model nor more data, but sorting people by *the hours at which they actually change location* and giving each kind of person its own set of trained numbers (Chapter 7).
2. **Population size stops paying at about two thousand people.** The edge grows from 0.6 to 4.2 correct answers per hundred between 400 and 1,900 training people, and a further ten-fold increase to 21,000 buys 0.1, with zero inside every confidence interval (Chapter 8).
3. **Difficulty lives in the person, not in the dataset.** 22.4 correct answers per hundred separate the stay-at-home user from the morning-moving user, against 0.4 across the entire range of training sizes: a factor of more than fifty (Chapters 5 and 8).

And one finding about measurement rather than modelling. The three largest single improvements of the nine weeks were not modelling ideas. Reading the best saved copy of a run instead of the last one was worth 15.5 correct answers per hundred. Recognising that where a day is cut in two belongs to the measurement and not to the model killed an apparent win. And choosing people on whom the one-line rule does not already score 69 % is what made the task measurable at all. On a problem whose honest yardstick is one line of code, the measurement deserves to be audited before the model is (Chapter 4).

Acknowledgements

My first thanks go to Robert Besták, who supervised this internship and took the time, week after week, to explain the parts of the mobile-network side of the problem that no amount of reading would have given me: what a signalling record actually is, how cells and base stations relate to one another, and why a question that looks like plain sequence prediction is constrained by the way the network itself is built. That patience shaped the direction of the work, and most of the corrections that mattered came out of those conversations.

I am equally grateful to the small group of interns working on the same corpus, Nino, Tiphain, Benjamin and Reda, for their help throughout these two months. They were already several months into their own analyses when I arrived, and they explained the corpus, the conventions and the pitfalls to me rather than leaving me to rediscover them. Nino in particular went out of his way, repeatedly, to produce the extracts I needed with the tooling he had built for his own analysis, which spared me a great deal of work and made most of the experiments in this report possible at all. Tiphain and Benjamin discussed their move-or-stay framing and their prompting results openly, and two of the ideas I reused here are theirs. Working next to them made those two months both easier and considerably more interesting than they would otherwise have been.

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1 Introduction

1.1 The host laboratory and its records

This report covers a second-year engineering internship carried out from 29 June to 28 August 2026 at the Czech Technical University in Prague (ČVUT), under the supervision of Robert Besták. The laboratory holds a large set of anonymous records donated by a Czech mobile operator, and hosts a small group of interns who study it from different angles.

Those records contain no names, no phone numbers and no message of any kind. One line says only this: an anonymous number standing for one person, and the list of antennas their phone was attached to during one day, with the time of each. That is the entire material of this internship.

1.2 The goal

A mobile phone is always attached to one antenna, the one that currently gives it the best signal. As the person walks, drives or takes a train, the phone lets go of one antenna and attaches to the next. The operator writes a line down each time that happens, and also at regular intervals when nothing happens at all.

The question I was given is short: *from the beginning of somebody's day, can I predict which antenna their phone will be attached to next?* An operator wants to know this because it decides where capacity is needed, and because a connection prepared before it is required is a connection that does not drop.

Two words are not synonyms, and the whole report depends on the difference.¹ A **base station** is a physical mast standing somewhere in the landscape. It carries several antennas pointing in different directions, and each of those is called a **cell**. The study area has **369 cells** on **113 base stations**, so a mast carries three or four cells on average. What I predict is a cell, not a mast, which is harder than predicting a place.

1.3 The instrument: a retrained language model

Figure 1.1 shows the whole internship in four steps, from the raw records to a ranked list of antennas, with the rule this work has to beat drawn underneath it. The instrument at the centre of that chain is a **language model**.² These are the programs behind the writing assistants everyone has now used: given the beginning of a sentence, they produce the word most likely to come next. Nothing in how they work is specific to words. They continue an ordered list of symbols drawn from a fixed set, and a day of movement is an ordered list of cells drawn from a fixed set of 369.

I do not use such a model as it comes. I **fine-tune** it: the numbers inside it are modified by my data until it stops being a general-purpose writer of English and becomes a predictor of

¹**Base station**, also called a *site* or a *mast*: one physical installation. **Cell**: the area covered by one antenna of that installation, pointing in one direction.

²**Language model**: a program trained to guess the next symbol of a sequence. Nothing in that description mentions language, which is why one can be pointed at a day of movement instead.

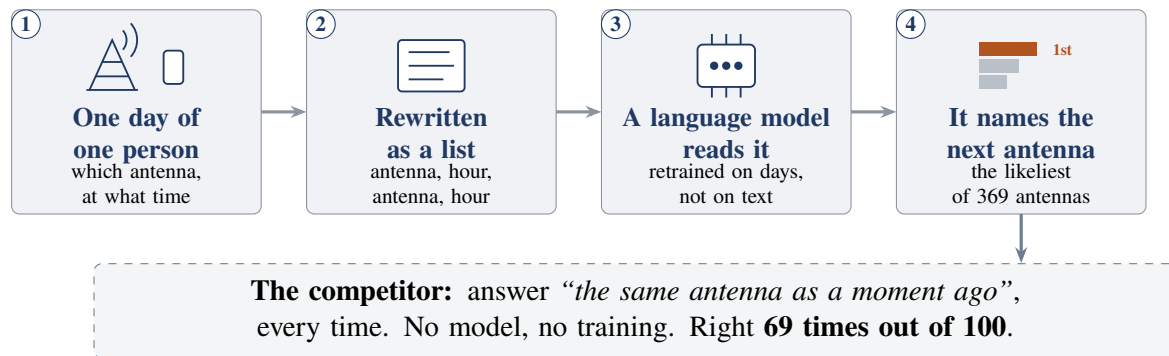


Figure 1.1: The four steps this internship built

What the picture says. The network records all day which antenna each phone is attached to. I turn one person’s day into a plain ordered list, retrain a small language model on thousands of those lists, and ask it to continue one: it ranks all 369 antennas of the area and names the likeliest. The dashed box is the real difficulty: a competitor with no model at all is already right about seven times out of ten, and every number in this report is the distance to it.

movement in one Czech region. Chapter 4 shows how. The other branch of the same problem, taken by two colleagues, leaves those numbers untouched and writes better instructions for a general model instead. That is called **prompting**, and it answers a different question rather than competing with mine.

1.4 The rule to beat: repeat the last antenna

What makes the question hard is not the model. It is the competitor.

Consider the rule “*the person is still on the same cell as a moment ago*”. No model, no training, no settings; one line of code. On an ordinary day of these records it is right **69.2 times out of a hundred**, simply because people spend most of their time not going anywhere. This rule is called the **persistence baseline** throughout the report, and beating it is the only result that means anything here. For five of the eleven iterations of this internship, nothing I trained managed to.

Figure 1.2 takes one prediction apart on two people, and it shows why the competitor is so strong. When the person stays put, repeating the last cell is correct and both the rule and my model collect the point for free. When the person leaves, the rule is guaranteed to be wrong, and the answer has to come from somewhere else. Every number in this document was won on moments of that second kind.

Every figure quoted from here on is a *difference between two percentages*. The model gaining 4.3 % over the rule means it answers correctly 54.5 times out of a hundred where the rule answers correctly 50.2 times, always on the same people and the same moments. Only that distance is comparable from one chapter to the next: the set of people I trained and tested on changed four times over the internship, each time because a new idea needed data the earlier files did not carry, and the difficulty of the test group changed with it.

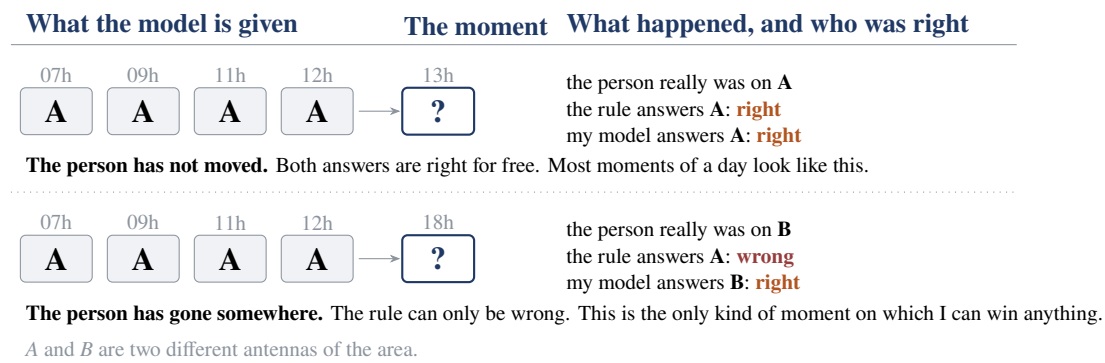


Figure 1.2: One prediction, taken apart

The two kinds of moment, and why only one of them counts. The rows are two moments of the same person’s day, and the letters stand for two antennas. On the left is the past the model may read, each moment an hour and an antenna; in the middle is the moment I want answered, hour given and antenna hidden; on the right, the rule’s answer and my model’s against what really happened. In the first row the person has not moved, so “the same as before” is correct and the rule scores a point without knowing anything. In the second the person has left, and the rule now has to be wrong, because being wrong is all it can do when somebody moves. That is the only kind of moment on which a model can gain anything.

1.5 The four colleagues working on the same records

Four colleagues were already working on the same records when I arrived, with different instruments. Figure 1.3 places my own branch among theirs. Nino was building his own description of the corpus: which cell is somebody’s home, which is where they spend their day, and who genuinely moves around the area. He had tooling for that, and he very kindly used it to produce the daily extracts and the user selections I asked for, none of which was his own assignment. Tiphain and Benjamin were taking the prompting branch, and Reda was working on the same records on a question of his own.

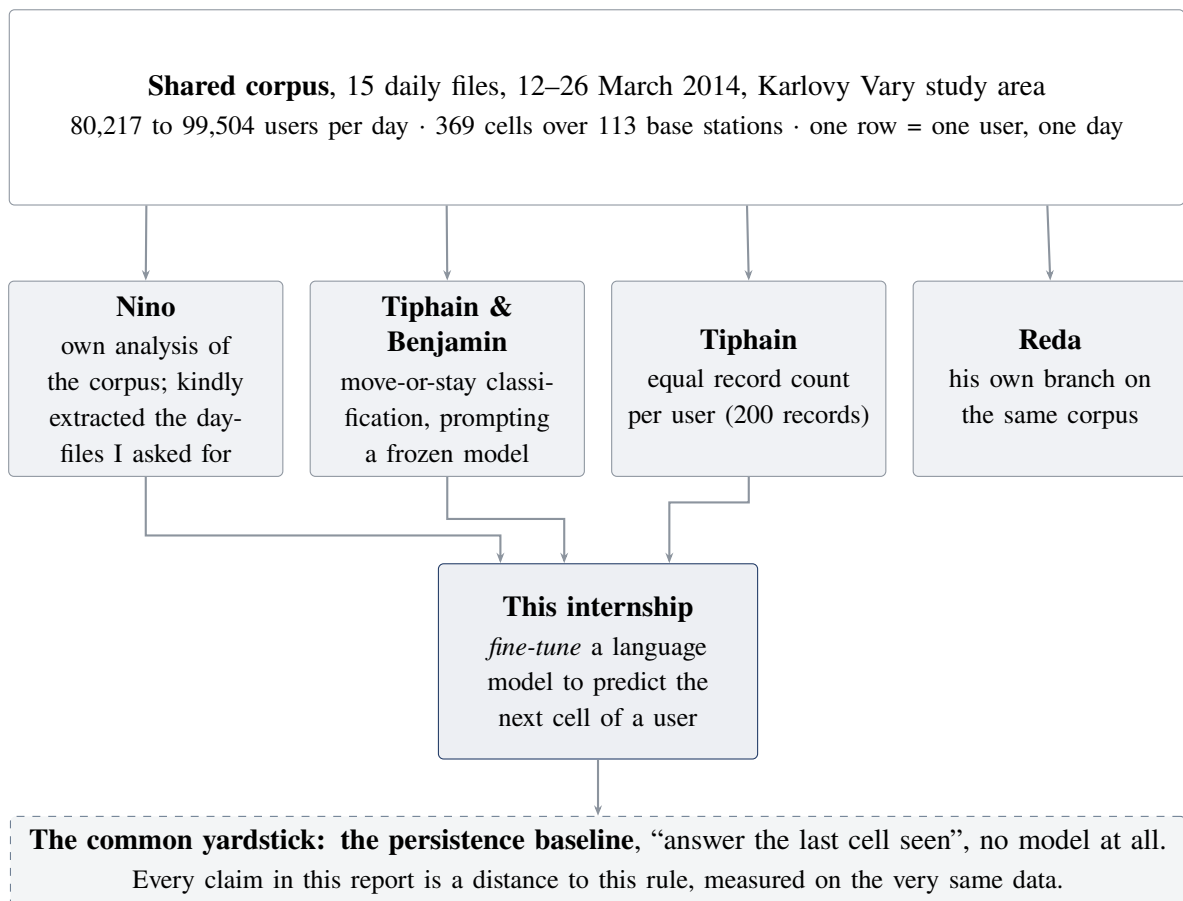


Figure 1.3: This internship inside the team’s work

Who did what, and why the dashed box matters. The four boxes across the middle are my colleagues. All five of us worked on the same fifteen days of records, drawn at the top, but on different questions and with different tools. Nino ran his own description of the corpus and, with the tooling he had built for it, extracted the daily files I asked for; Tiphain and Benjamin worked on the prompting side and lent me two ideas I reused; Reda worked on his own question. My own box is at the bottom. Different tools produce numbers that cannot be compared unless everyone is graded against the same thing, and the dashed box is that common thing: a rule with no model in it, “answer the last cell seen”.

1.6 The plan of this report

The chapters are organised by *question*, not by week; the chronology is restored in Appendix A. Each of Chapters 3 to 8 closes with a *Conclusion* section, written after the fact, that states what the chapter established and what I make of it with hindsight. Appendix B gives the settings of every run and the origin of every headline number, and Appendix C defines the terms that recur.

2 State of the art

Eight published works matter to this report, and all eight are listed in the bibliography at the end. Two of them changed the way I work and are worth the first section on their own. Of the six that follow, three settle how a predictor of this kind is normally built and normally scored, two are the ancestry of my only substantial gain, and the last one marks the boundary between the published task and the one I actually took on.

2.1 The narrow room a predictor has

The most useful part of this literature is not the part that proposes predictors. It is the part that measures how much room a predictor has before it even starts. As Figure 2.1 shows, once the published ceiling is corrected for people simply standing still it falls from 93 to 71 correct answers per hundred, and the one-line rule of this report already sits at 69.2 of them: read on that single scale, almost nothing is left to win.

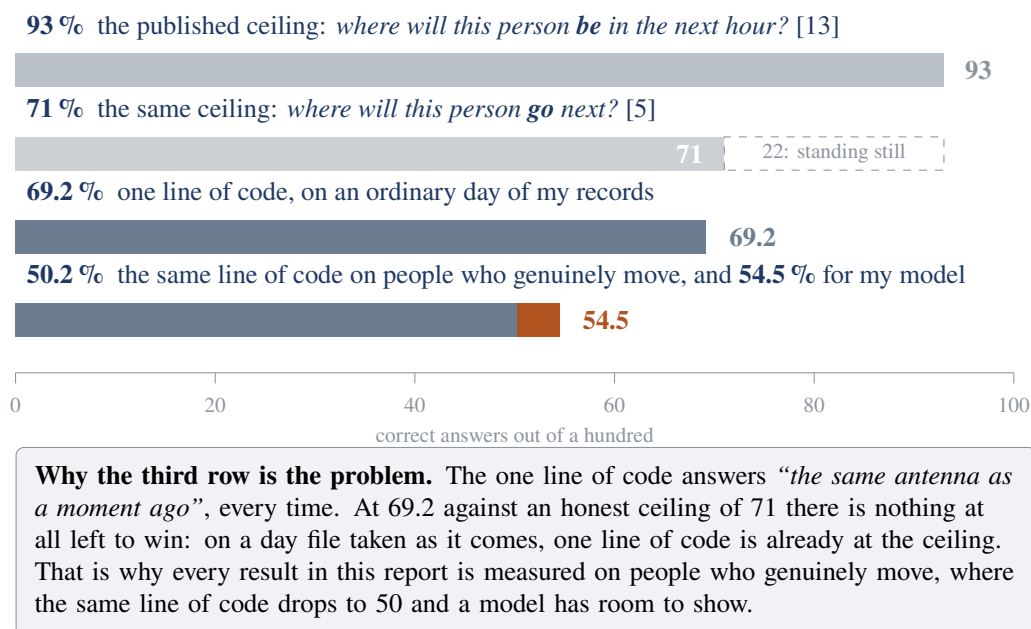


Figure 2.1: The room a predictor has, once standing still is taken out of it

Reading the four bars. All four are on one scale, the share of moments answered correctly. The two grey bars are published ceilings, the best score any predictor could ever reach; the two blue ones are what the trivial rule and my model score on my data. Song et al. put the ceiling at 93 %. Ikanovic and Mollgaard re-ran the same measurement after deleting the moments where the person had not moved, and it fell to 71 %, the dashed part of the second bar.

The two bottom bars are different sets of people, deliberately. An ordinary day file and a selection of people who genuinely move are not comparable on absolute score. The rule scores 69.2 on the first and 50.2 on the second without changing a character of its own definition; Chapter 3 makes that move and explains it.

Song et al. took three months of mobile-phone records for 45,000 people, treated each person’s sequence of visited antennas as a stream of symbols, and asked how much of the

next position is already implied by everything that came before [13]. From that they derived a *ceiling*, the best score any predictor could ever reach, however clever: 93 %.

Ikanovic and Mollgaard then showed that most of the 93 is not knowledge about anybody [5]. The quantity Song et al. bound answers “*where will this person be during the next time slot*”, and in most time slots the answer is “exactly where they are now”. Re-running the same machinery on the question a predictor is actually useful for, “*where will this person go next*”, with the repetitions removed, the ceiling fell to 71 %. Roughly a quarter of what looked like predictability was the passage of time rather than the behaviour of a person. That is what names my competitor: if most of a high score is people staying where they are, any honest claim has to be a distance to “answer the last antenna seen” rather than an absolute score.

Two further things in Song et al. bear directly on later chapters.

The ceiling is the same for everybody, but the score is not. The authors expected heavy travellers to be harder to predict and were wrong: past a radius of ten kilometres, the ceiling stops depending on distance travelled at all. Yet what can be predicted from a person’s visit frequencies alone is spread very widely across the population and peaks near 30 %. A population average therefore hides the thing worth reporting, which is why Chapters 5 and 7 measure the spread directly.

And a curve published almost in passing, which cost me an iteration. Splitting the week into 168 hourly slots, the authors measure how often a person is found in their most-visited place for that hour. The answer is close to 90 % during the night, with clear low points around midday and between six and seven in the evening. The hours at which people are hard to predict were published in 2010, and they are not the small hours of the morning. Chapter 6 spends its whole length rediscovering that the expensive way.

2.2 The contribution of the six other cited works

The first three of the six settle how a predictor of this kind is normally built, and how its answers are normally scored. Luca et al. survey the deep-learning branch of human mobility [8], and two of their observations are used without modification here: giving a model the time of day is standard practice, so the mechanism of Chapter 6 is not an idea of mine, and scoring a prediction at three depths, top-1, top-3 and top-5, is the convention of the field rather than a choice of mine. The language-model work closest to this one reports the same three depths [11]. The convention exists for a practical reason: an operator that can prepare three antennas instead of one is served by a correct answer in third place. The same survey notes that the better step-by-step networks had to be extended with a mechanism for reaching back to a matching moment in an earlier day. That reaching back is the native operation of the architecture behind current language models,¹ which is one honest reason to reach for an existing language model here rather than build a predictor from scratch.

Since 2023 the field has split into the same two branches the team I joined was split into. Wang et al. leave the model’s numbers alone and write a better instruction [14]: that is the branch I did not take, and in substance what Tiphain and Benjamin were doing on this corpus. Qin et al. change the numbers instead, with the same cheap method used here, over

¹**Transformer:** the architecture behind current language models, in which every position of a sequence can look directly at every earlier one.

six mobility datasets [11]; of the eight, that is the work mine is closest to.

The last two of the six are the ancestry of Chapter 7, and I found them only after the fact. Pappalardo et al. establish that a population separates into recognisable kinds of person rather than spreading smoothly along one scale of mobility [9]. Jeong, Leconte and Proutière supply the mechanism: one person's history is too short to estimate that person's habits from, so group people who resemble each other and predict one person's next location partly from the others' [6]. My four kinds of user are the same idea with the axis changed, from how far a person travels to at which hours they change location.

2.3 Predicting an antenna, not a place

Everything above predicts a *place*: a home, a workplace, a stopping point obtained by grouping nearby records. My target is the next antenna, which is a different object. The work that predicts antennas sits in the networking community and asks the question for another purpose: Chen, Mériaux and Valentin predict which antenna a phone will be handed over to,² from radio measurements, so the network can prepare it [1].

Predicting an antenna from a trace of antennas, with no radio measurements, inherits a drawback from each side. In this corpus 12.4 % of back-to-back records change antenna without the person having gone anywhere, because the phone re-attached to another antenna of the same mast (Chapter 3), so part of what my model is asked to predict is a decision of the network rather than of a human being. And the human part of it is graded on antennas rather than on places, which is a finer target than the published ceilings were ever estimated for.

²**Handover:** the moment the network moves an ongoing connection from one antenna to another, normally because the new one offers a better signal.

3 The data

3.1 The goal

Before anything can be predicted, one has to know what the data contains. Two answers shaped every experiment that follows: **a record is not a movement**, and **the name of an antenna is not a place**. Both are the reason several later results look surprising at first sight.

3.2 One line of the data, taken apart

The material is fifteen daily files covering 12 to 26 March 2014 in the Karlovy Vary study area of north-west Bohemia. They are plain text files¹ in which **one line is one person for one day**, from midnight to midnight. Depending on the day a file holds between 80,217 and 99,504 lines; the day used for most of the work here, 12 March 2014, holds 85,946 people.

A line starts with eight fixed columns, then repeats a pair of values for as many records as that person produced. The eight are an anonymous number standing for the person, their age, their gender, an unexplained five-digit number, a behaviour class, a “home antenna”, an “activity antenna” and the number of records that follow. Then, over and over, **the name of an antenna followed by a time**, written as a number of seconds since midnight from 0 to 86,399. Only the anonymous number, the record count and the antenna-and-time pairs are used here: the rest carry known aberrations, including ages of 1078 and negative values.

A record is written whenever a phone and a mast exchange a message, and that includes the case where nothing happens at all: a mast asks every phone attached to it to announce itself on a fixed cycle, typically every four hours on 2G and every two on 3G,² so that it knows where to route an incoming call. So **a record is not a movement**: somebody who never leaves their flat still produces records all day long, and how many they produce says as much about which generation of radio their phone attached to as about what they did. A long run of one antenna name is also what makes “repeat the last name you saw” such a hard competitor.

3.2.1 The rule to beat, and the reason its value moves

The **persistence baseline** is the rule “the next antenna is the last one seen”. No model, no training, no settings, and it is scored on exactly the same moments as my model, in exactly the same way.

Its value is not a constant, and this is the part that is easy to miss. It is a property of the people being scored. On a day file taken as it comes, where a large share of people barely move, it is right 69.2 % of the time. Once the data is restricted to people who actually move around, the very same rule falls to 45.5 %. Nothing about the rule changed; the population did.

¹**CSV, comma-separated values**: the simplest tabular text format, one row per line with the fields separated by a delimiter. Here the delimiter is a semicolon.

²**Re-signalling, or periodic location update**: the network asking every attached phone to announce itself on a fixed cycle, so that it knows where to route an incoming call.

3.3 The name of an antenna is not a place

An antenna name such as BKVCH1 looks like an opaque location code. It is not, and this is the point at which the technical name for it can safely be introduced: in the working files this string is called the **cell_id**, and it is used with that name from here on.

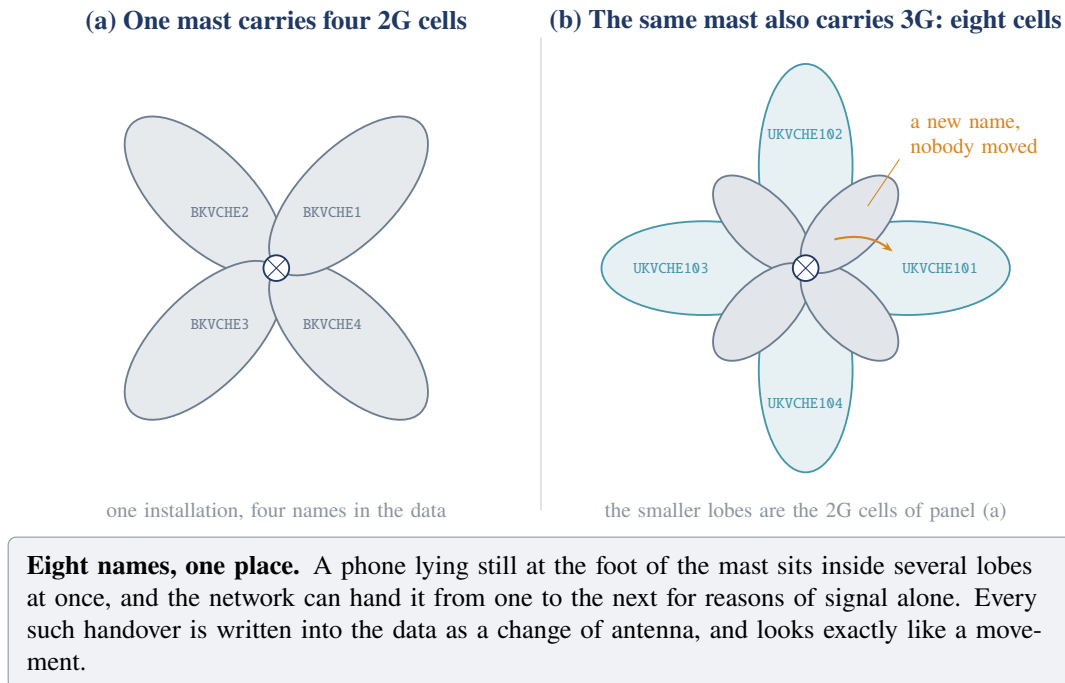


Figure 3.1: One mast, and all the names it answers to

What is drawn, and why it decides how movement is measured. Both panels are the same mast of the study area seen from above, the crossed circle being the installation itself and each shaded lobe the area covered by one of its antennas pointing in one direction. Panel (a) has the four 2G cells alone; panel (b) adds the four 3G ones, which reach differently and point elsewhere. The eight lobes all start at the same point, so any hop among those eight names happens at one address. On the day used for Chapter 7 that is **12.4 out of every 100** back-to-back records.

Figure 3.1 draws one real mast with every name it answers to, and the count underneath it is the whole point: eight names, one place. One mast answers to several names at once, one per generation of radio and one per direction its antennas point in, so there are three or four times as many names as there are masts. A phone that never leaves the same spot can be passed from one of those names to the next, and in the raw data that looks exactly like somebody going somewhere.

Figure 3.2 then counts how often the data plays that trick: on this corpus 12.4 of every 100 back-to-back records change the name of the antenna without the person having gone anywhere. So whenever this report asks whether somebody *moved*, it looks at the mast and not at the antenna name. What the model has to *predict* stays the full antenna name, because that is the question I was given; predicting only the mast would be an easier one.

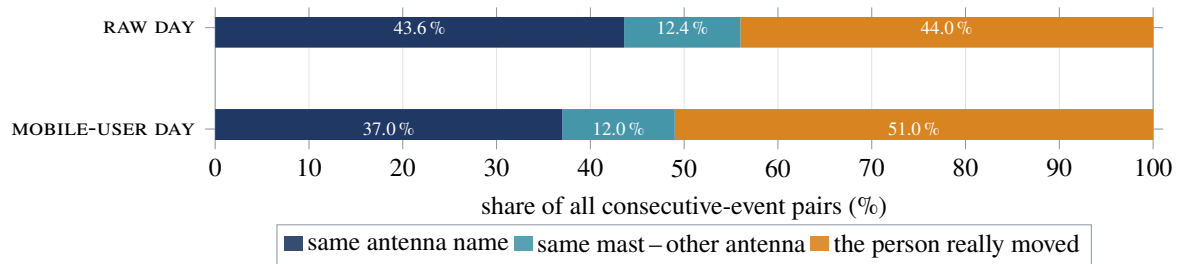


Figure 3.2: The three things a pair of back-to-back records can be

What is being counted. Take every pair of back-to-back records in a file and ask what separates the two. Each bar is 100 % of one file’s pairs, cut three ways. Dark blue: the same antenna name, so nothing at all happened. Teal: the antenna name changed but the mast did not, so the phone hopped to another antenna of the same installation while the person stayed where they were. Orange: the mast changed too, so the person genuinely went somewhere.

The middle band is the trap, and filtering cannot remove it. On an ordinary day only 44.0 pairs out of a hundred are a real movement, and 12.4 are the middle band: they look like movement in the raw data and are not. Once only mobile people are kept, real movement rises from 44 to 51, while the middle band stays at 12. That band is a property of the radio network, not of the people, so mobility is always measured on the mast. The prediction target deliberately stays the full antenna name, and the price of that is accepted openly (Chapter 5).

3.4 From a raw file to training material

Figure 3.3 follows one day file through the four steps that turn it into material a model can be trained on. Two choices inside that pipeline change how later results should be read.

The split is by person, and it is balanced. Training and test people are different people: the model is graded only on strangers. The assignment is not random but *balanced* on the number of records per person,³ because without that a split can silently hand all the longer histories to one side.

Cleaning removes people, not records. Three filtering decisions were made over the internship, each in response to a result rather than in advance. The earliest runs used the day file exactly as it came, which is why the rule sat at 69.2 %: a large share of people produce a handful of records from a single antenna, so the evaluation was dominated by moments where there was nothing to predict. Then people with ten records or fewer, and people with a hole longer than four hours, were removed; four hours is the corpus’s own convention, since a mast re-signals at least that often, so a longer gap means the person left the recording area. Finally the data was restricted to people who move and stay in the area, using the per-day selection prepared by a colleague. Each step makes the *task* harder and the *rule* weaker, which is exactly the point: a benchmark on which one line of code already scores 69 % leaves almost nothing to measure.

³**Stratified split:** people are first sorted into bands by how many records they have, and the split is made inside each band.

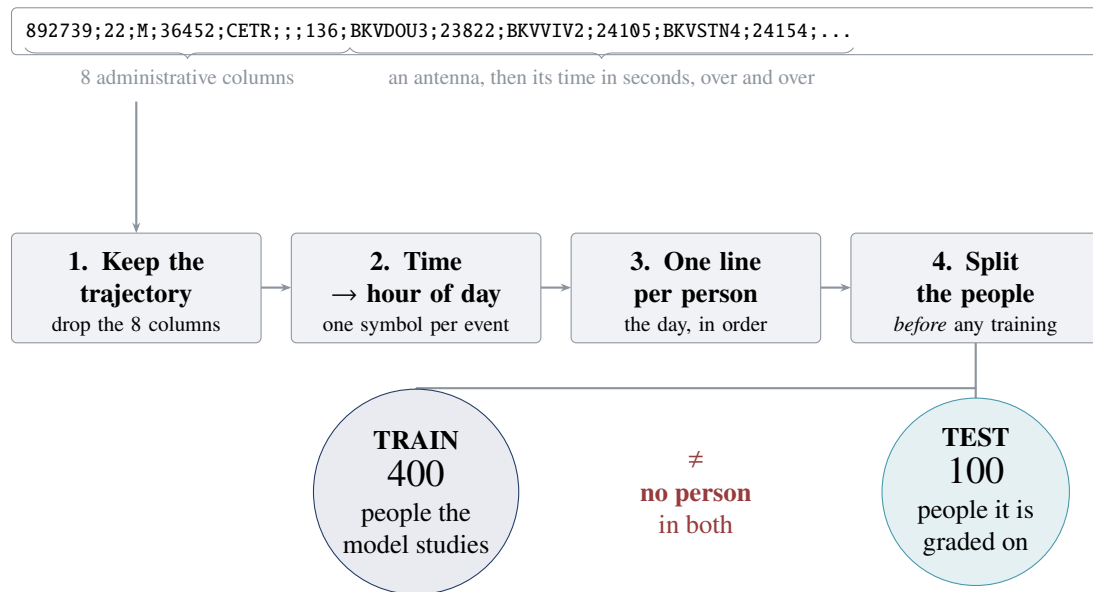


Figure 3.3: From a raw file to training material, in four steps

The raw line, and the four steps. At the top is one person's whole day as it arrives from the operator: eight administrative fields, then the name of an antenna and the time in seconds since midnight, over and over. **(1)** Throw away everything that is not a trajectory. **(2)** Turn each time into the hour of day it falls in. **(3)** Write one line per person, their day as an ordered list of events. **(4)** Split the people into those the model studies and those it is graded on. The two circles hold different people, so nothing the model is scored on can have been memorised, and the split is balanced so that both sides get the same mix of short and long histories.

3.4.1 The six sets of people used in this report

Table 3.1 lists all six, with the score of the one-line rule on each in the last column. Six sets for one problem is the direct consequence of each new idea needing data the earlier files did not carry. Adding the time of day required regenerating the sequences with their times kept. Testing whether more people help required a day file with more people in it. Testing whether *many* more help required stacking eleven day files end to end.⁴ Each regeneration produced a new test group, and therefore a new value for the rule.

⁴**Stacking** day files: putting them one after another into a single training file, which adds *people* rather than adding days to the people already there.

Table 3.1: The six sets of people, and the score of the one-line rule on each

	Who is in it	Train	Test	Records/person	Rule
RAW DAY	one day file taken as it comes (12 March 2014)	400	100	mean ≈ 62	69.2 %
FILTERED DAY	the same day, filtered: more than 10 records and no 4-hour hole	400	100	>10 by design	79.0 %
MOBILE-USER DAY	people who move, and who stay in the area, on one day	400	100	19–200, mean 86.0	45.5 / 43.1 %
LEVELLED CORPUS	the same people, every day stretched to exactly 200 records	400	100	200 exactly	76.5 %
2,000-USER DAY	a day file sampled down to 2,000 people	900 / 1,900	100	12–200, mean 77.1	48.2 %
ELEVEN-DAY SET	11 day files of 2,000 people, stacked end to end	21,000	1,000	10–200	50.2 %

How to read the table. One row per set of people used over the eleven iterations. **Train** is how many people the model studied and **Test** how many it was graded on, always different people. **Records/person** says how long a typical day is in that set. The last column is the score of the one-line rule on that set's test people, and it is the number every result in the corresponding chapter is compared against.

Why the last column moves so much. It is a property of the people, not of the rule. Nothing about “repeat the last antenna” changes between 69.2 and 43.1 %; what changes is who is being scored. This is the single reason an accuracy from one chapter must never be compared with an accuracy from another. **MOBILE-USER DAY** carries two values because 45.5 % is measured on the last fifth of each test person's day and 43.1 % on every position of it, one at a time, after the evaluation was widened in the sixth iteration; scoring every position is strictly harder.

3.5 The questions this data cannot answer

The unit is one day. The question posed at the start of this internship is a within-day one: predict the rest of a person's day from its beginning. Every experiment here answers it. Predicting tomorrow from the days before is a different question, and the natural continuation of this work.

The world is small. 369 antennas over 113 masts, with a typical person visiting on the order of twenty distinct ones. Already at the smallest training size, 98.9 % of the antennas appearing in the test people's days had also appeared in the training people's, and from 5,000 people onward it is 100 %. So discovering an unseen antenna cannot be a source of improvement beyond that point: any remaining gain has to come from better estimates of the moves between antennas already known.

3.6 Conclusion

Two properties of this material constrain every result that follows. A record is not a movement: the network writes a line whenever it asks a phone to announce itself, so a day of records is largely somebody sitting still being counted over and over, and that is exactly what makes

“repeat the last antenna” so hard to beat. And the name of an antenna is not a place: one mast answers to three or four names at once, so 12.4 of every 100 changes of name are changes of radio rather than changes of location. From here on, whenever this report says that somebody *moved*, it has looked at the mast; what the model is asked to produce stays the full antenna name, because that is the question that was put to me.

The decision that mattered most in this chapter was not a modelling one. It was accepting that the raw day file could not support the question being asked: moving to the selection of people who genuinely move around the area cost the comfortable-looking accuracies of the early runs, and that is precisely what made every later chapter measurable. With hindsight the order was wrong. I spent the first iterations improving the training machinery on a file where one line of code already scored 69.2 %, and only then looked at what was in it; a day spent reading the data first would have shown that there was nothing to win there. The price of changing it late is the six sets of people of Table 3.1: the yardstick moves from one chapter to the next, so no absolute score in this report may be compared with an absolute score in another. Only the distance to the rule travels.

4 The training procedure and the measurement rules

4.1 The goal

Two practical questions had to be settled before any experiment could mean anything. *Given one free graphics card and a few weeks, which model and which training method are actually feasible? And what has to be true of the measurement before a trained model's score can be believed?*

The answer to the first was decided in the first week and never changed. What changed over the internship was the data and the training signal, never the instrument, and that is what makes the later comparisons between runs meaningful. The answer to the second cost me three wrong conclusions, one per rule, and the second half of this chapter is those three rules.

Where the numbers in this chapter come from. From the three earliest sets of people of Table 3.1: the `RAW DAY` taken exactly as the operator sent it, the `FILTERED DAY` once people with too few records were removed, and the `MOBILE-USER DAY` once only people who genuinely move are kept. Each uses **400 people for training** and **100 different people**, never seen during training, for grading. The one-line rule scores 69.2 %, 79.0 % and 45.5 % on them respectively, and that difference between the three is itself the subject of the third rule below.

4.2 The tools, and the share of the model that is changed

A language model is a stack of very large tables of numbers.¹ Adapting one somebody else has already trained² can be done at three very different costs, and what separates them is not accuracy. It is memory.

Figure 4.1 puts the three costs side by side: retraining everything needs far more memory than a free card has, LoRA freezes the original numbers and trains a small set of new ones beside them, and QLoRA does the same after shrinking the frozen numbers to a coarser precision. I chose QLoRA, and on hardware grounds rather than on expected quality: it was the only one of the three under which my first candidate model could be loaded at all on a free graphics card. That choice held for the rest of the internship, and it is what made the volume of experiments possible, and the per-group routing of Chapter 7 cheap enough to try at all.

4.2.1 The choice of model

The first real run used **Mistral-7B** [7], a general-purpose model with seven billion numbers in it.³ A deliberately small first experiment, about thirty minutes over 1.2 % of the training

¹**Weights**, or **parameters**: the numbers inside the model that turn an input into an output. “Frozen” means left exactly as its original authors trained them.

²**Pre-trained**: already trained, by someone else, on a very large body of general text. Fine-tuning starts from those numbers rather than from nothing, which is what makes it affordable.

³“7B” means seven billion numbers. Qwen2.5-0.5B has five hundred million, fourteen times fewer. The count is the model's capacity, and also how much memory it occupies.

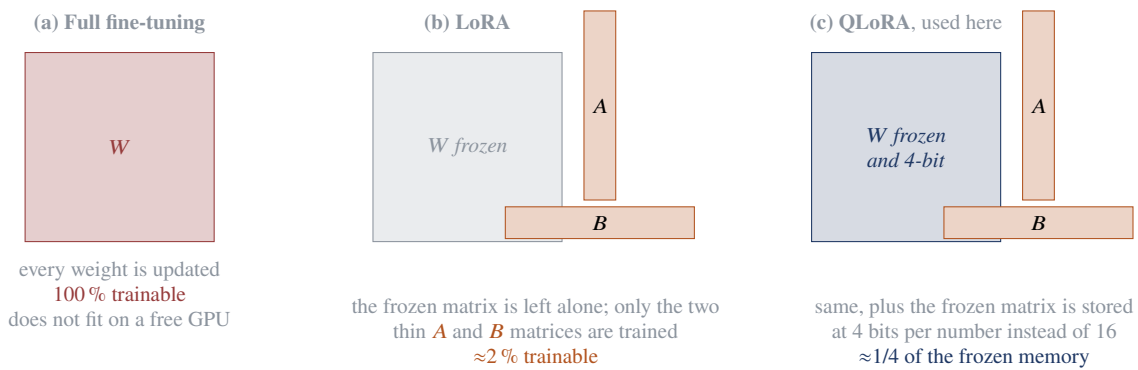


Figure 4.1: Three ways to adapt a model, and their cost in memory

Three ways to adapt a model, at the same scale. The grey square is the model, a stack of very large tables of numbers. In (a) every number is modified, which has the highest ceiling in principle but requires holding each number together with about three more used to track how to adjust it, and it does not fit on a free card. LoRA (b) [4] freezes the big table and trains two narrow ones added beside it, so about 2 numbers in every 100 ever move. QLoRA (c) [2], used throughout this report, does the same and stores the frozen table with four bits per number instead of sixteen, which is safe precisely because those numbers are read and never updated. What this decides is not accuracy but waiting time: one run in half an hour rather than five and a half.

data, produced 16.87 % correct next-antenna answers, which was encouraging as a signal that the pipeline worked end to end. A Llama model tried at the same size did not even return well-formed antenna names.

Scaled up to a full run on 400 training people, however, Mistral-7B spent roughly five and a half hours on the free card to reach a best score of 41.1 % and then stopped improving. The same data given to **Qwen2.5-0.5B-Instruct** [12], fourteen times smaller, ran forty-nine passes over the data in about seventy-two minutes, and twenty-nine minutes once the rules below were in. Mistral-7B was abandoned.

The reasoning, stated at the time and still my view, is that the task is narrow. Naming the next antenna out of 369 requires no knowledge of the world and no reasoning in English; it requires learning how one small geographical area is moved through. The larger model brings capacity the task cannot use and charges for it in waiting time, by a factor of roughly twenty-three per run, which over nine weeks is the difference between thirty experiments and two.

Figure 4.2 gathers the whole toolchain in four layers, from the free graphics card at the bottom to my own notebook at the top; everything between the two is somebody else's library.

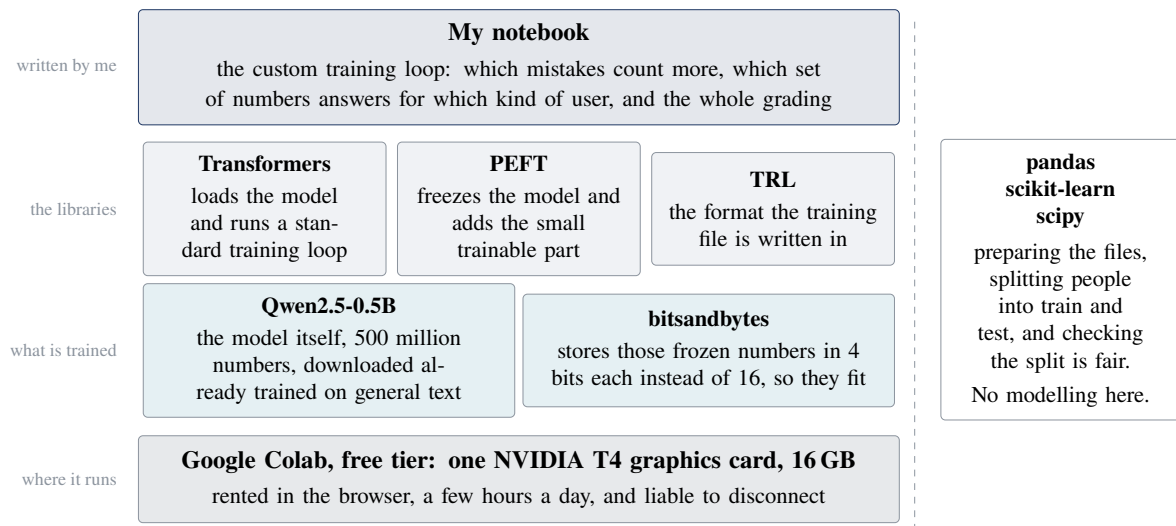


Figure 4.2: The tools, and the job of each one

Read the stack from the bottom up. At the bottom is the machine, a graphics card rented for free through a browser, which is the constraint behind almost every choice in this chapter. Above it sit the model and the trick that makes it fit in that card's memory, then three off-the-shelf libraries: **Transformers** loads a published model and runs a standard training loop, **PEFT** makes training affordable by freezing the model's 500 million numbers and adding a small set beside them, and **TRL** contributes only the format its files are written in. The column on the right never touches the model.

Why the top layer had to be mine. Everything this report is about lives there. Making a mistake at seven in the evening count more than one at two in the afternoon, and sending one kind of user to one set of numbers and another kind to another, cannot be expressed by changing a setting. Both require reaching inside the training step, and that is the single property the whole stack was chosen for in the first week.

4.3 Interpretation of data and implementation of model

The model cannot read a day; it reads symbols. Figure 4.3 shows one person's day exactly as the model receives it.

Three choices are visible in that picture. **One antenna, one symbol:** all 369 antenna names were added to the model's vocabulary⁴ as symbols of their own, since left alone the tokenizer would have split BSOCH01 into four meaningless fragments and the model would have had to spell an antenna correctly to be right. **One hour, one symbol:** from the sixth iteration each event is preceded by a symbol for its hour of day, <H00> to <H23>, which is twenty-four learnable symbols. And **only the antennas are graded:** the opening boilerplate and the hour symbols are excluded from the score,⁵ so the model is trained, and graded, only where an antenna name has to be produced.

(a) One person’s day, as it is stored on disk. One line of the training file, spread over several lines here so that it can be read. Nothing is a conversation in any real sense: the three labels are simply the slots the training library expects to find.

```
{
  "conversations": [
    {
      "role": "system",
      "content": "You are an AI that generates a user's cell-visit trajectory."
    },
    {
      "role": "user",
      "content": "Here is the sequence for this user."
    },
    {
      "role": "assistant",
      "content": "User 2832452 | 200 events | BSOCHO1 BSOCHO1 UKVCHE102 BSOCHO1 ..."
    }
  ]
}
```

system	What the task is. The same words on every line of the file.
user	The request. Also the same everywhere: a placeholder, not a question.
assistant	The person’s actual day. This is what the model has to learn to produce, and the only part its mistakes are counted on.

(b) The same line, once it has been cut into symbols. This is what the model actually reads. Each antenna name is *one* symbol, not a spelling; each event is preceded by its hour. Only the dark blue positions are graded.

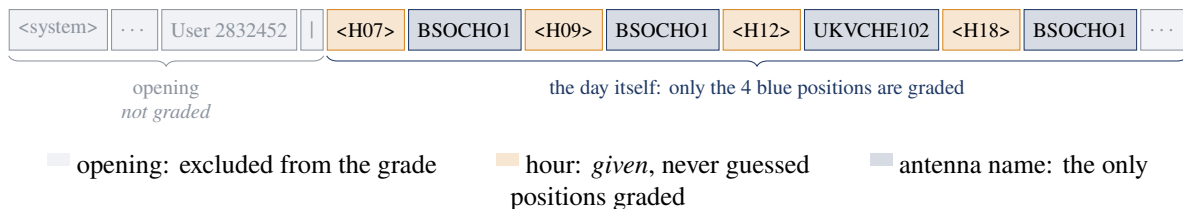


Figure 4.3: One person’s day, as the model receives it

Panel (a) is the file on disk; panel (b) is what the model reads. Each person’s day is written as a record with three slots, two of them boilerplate and the trajectory in the third, a layout chosen only for compatibility with a library that already knows how to read it. In panel (b) the grey symbols set the scene and are excluded from the grade; then the day arrives as pairs, an orange symbol for the hour and a dark blue symbol for the antenna. Each antenna name is a single symbol rather than a spelling, so the model chooses among 369 known answers instead of spelling one letter by letter. The hour is handed over for free and never guessed, and it costs one symbol rather than five: writing “07:31:22” would have cut the visible history by roughly two thirds.

4.3.1 One training run, step by step

Figure 4.4 sets out the five steps one run goes through, and the rule underneath them that decides which saved copy is kept. Two further mechanisms were added to this loop from the sixth iteration onward, and later chapters depend on both. **Some mistakes are made to count more:** at step 3 the surprise measured at a given position can be multiplied according to the hour that position falls in, so a mistake at seven in the evening under a multiplier of four costs four times the same mistake at two in the afternoon. No data is duplicated; the optimiser is simply told which mistakes matter. **Some slices are shown more often:** at step 1, slices containing an interesting moment can be drawn three times more often than the rest, which

⁴**Vocabulary:** the fixed set of symbols a language model can read and produce. **Tokenizer:** the component that cuts an input into those symbols.

⁵**Masking** a position means excluding it from the score: the model still *reads* the symbol but is neither rewarded nor penalised for producing it.

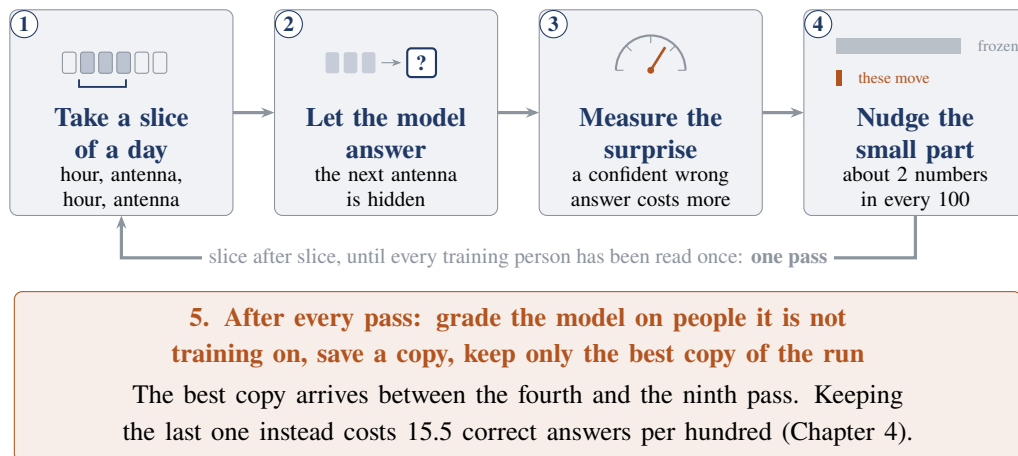


Figure 4.4: The five steps of one training run

Steps 1 to 4 are the whole of training. A slice of somebody's day is shown to the model with the next antenna hidden. The model answers, the answer is compared with the truth and turned into a single number measuring how surprised the model was, and that number nudges the model's adjustable numbers in the direction that would have made it less wrong. Only a small part ever moves: the 500 million numbers the model was published with are frozen, and the nudging applies to about 2 in every 100 added beside them.

Step 5 is not an optimisation, it is a safety rule. Left to itself, a model trained on a few hundred short days stops learning the habits of the population and starts learning the days by heart, while the training figures happily report success for another forty passes. Grading on separate people after each pass, and keeping the best copy rather than the last, was the largest single improvement of the whole internship.

changes how often the question is asked rather than how much a wrong answer costs. Both turned out to help only where the targeted positions were genuinely the hard ones, which is the subject of Chapters 6 and 7.

4.3.2 The machine: one free graphics card

Every run reported in this document was trained on **Google Colab**⁶ on an NVIDIA **T4**. That card carries 16 GB of memory, and that single figure is the constraint behind every choice in this chapter. The daily free quota is why the experiment schedule is one training per iteration rather than broad sweeps of settings.

Three standard libraries do work that never touches the model.⁷ They split people between training and test, check that the two halves have comparable histories, and run the resampling test [3]⁸ that decides the central question of Chapter 8.

⁶**Google Colab**: a service that runs a Python notebook on a graphics card rented in the browser, free for a few hours a day and liable to disconnect. **T4** is the entry-level NVIDIA card it hands out.

⁷**pandas**, **scikit-learn** and **scipy**: the three standard Python libraries for, respectively, manipulating tables of data, machine-learning utilities, and statistical tests.

⁸**Bootstrap**: error bars obtained by re-scoring the models on many random redraws of the test people, with no assumption about the shape of the underlying distribution.

4.3.3 The pieces reused from the team

Three pieces came from the team. The daily extracts and the selection of people who move and stay in the area came out of Nino's own analysis tooling, and those files are the data on which most of this report's modelling rests. The move-or-stay framing is Tiphain and Benjamin's, and it is the idea behind the filtering of Chapter 3 and behind reporting in-window and out-of-window scores separately. The technique of levelling every person's record count, tested in Chapter 8, is Tiphain's. Their branch of the problem sends one long instruction, narrating a day in English, to a model whose numbers are never modified: everything it is meant to know about this population has to be restated in every instruction, whereas the branch this report takes stores that in the numbers instead.

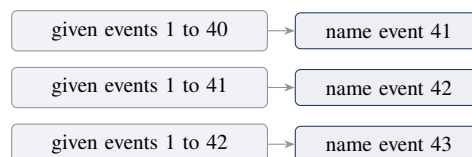
4.4 The scoring procedure

Everything in this report is scored the same way, and the procedure never changes. Each test person's day is cut once into a past the model may read and a rest it has to produce, the default being 80/20. The rest is then answered one moment at a time: the model is asked for one antenna, told what really happened, and asked for the next, so it can neither see the future nor compound its own mistakes. The one-line rule is run through exactly the same moments with exactly the same history. Figure 4.5 draws that procedure on one day, with the cut, the part the model may read and the part it has to produce.

(a) Each person's day is cut in two, once and for all



(b) The rest is then answered one moment at a time, never in one go



each time the real past, never the model's own earlier answers

Figure 4.5: The cut of the day, and the answering of each moment

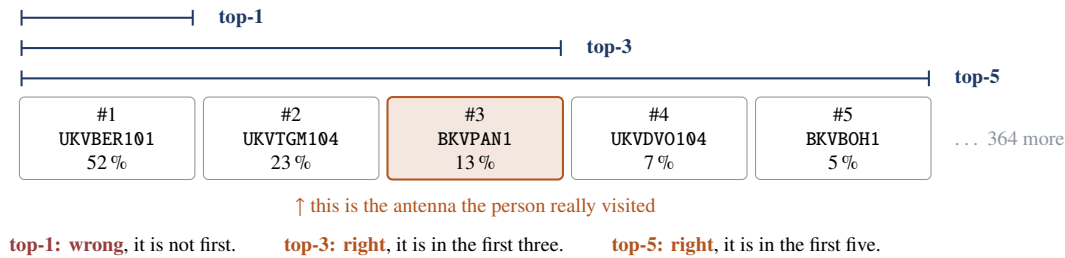
Panel (a) is the cut; panel (b) is how each moment is answered. Each test person's day is split once into a part the model may read and a part it has to produce, the default being 80/20. Where that cut falls is part of the *measurement* rather than a property of the model. The model is then not asked to invent a whole afternoon: it is asked for one antenna, told what really happened, and asked for the next, so it can neither see the future nor compound its own errors. The one-line rule is run through exactly the same moments with exactly the same history, which is what makes the two numbers comparable at all.

At each graded moment the model produces not a single answer but a ranking of all 369 antennas, most likely first, and the moment counts as correct if the antenna the person really visited is at the top of that ranking. Reading the ranking one place down gives **top-3**, and further down **top-5**. Figure 4.6 shows the same ranking read at those three depths, and its

lower panel shows how a score is obtained from a row of right and wrong answers.

(a) The model never answers once. It ranks all 369 antennas.

Here are its first five, for one moment of one person's day, with how sure it is of each.



(b) The score is then just a count, over every graded moment of every test person.

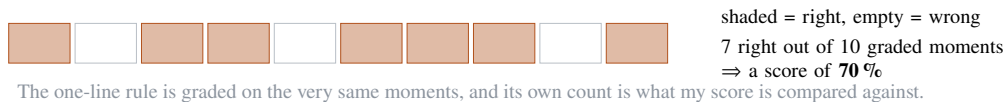


Figure 4.6: Top-1, top-3 and top-5, and the way a score is obtained

Panel (a): the model does not give an answer, it gives an order. Asked what comes next, it assigns a degree of confidence to all 369 antennas and sorts them, most likely first: here 52 % for UKVBER101, 23 % for UKVTGM104, and so on. **Top-1** asks whether the right answer is in first place, **top-3** whether it is anywhere in the first three, **top-5** in the first five. In the example the right answer sits third, so the same prediction fails at top-1 and succeeds at the other two. Reporting three depths is the convention of this literature rather than a choice of mine [8, 11].

Panel (b): where a percentage comes from. The score is a plain count. Every graded moment is answered, marked right or wrong at the chosen depth, and the score is the number of right answers over the number of moments: ten moments, seven right, 70 %. The one-line rule answers the very same moments and gets its own count, and every result in this report is the difference between the two.

The three depths are not three models; they are the same predictions read at three levels of strictness. Top-1 is the strict headline used throughout this report; the other two say whether a model that misses is missing narrowly or wildly.

4.5 Three rules about measuring

Figure 4.7 states the three rules and what each one was worth, and the rest of this section establishes them one at a time.

4.5.1 Rule 1: never read the last pass of a run

A language model fine-tuned on a few hundred short days will end up learning them by heart.⁹ That is not a hypothesis; it is what the curves do.

As Figure 4.8 shows, on four different sets of people alike the score on strangers rises for a few passes, peaks, and then falls for the rest of the run. The clearest evidence is a single

⁹**Overfitting:** the point past which a model stops learning the regularity of the data and starts memorising the examples it was shown.

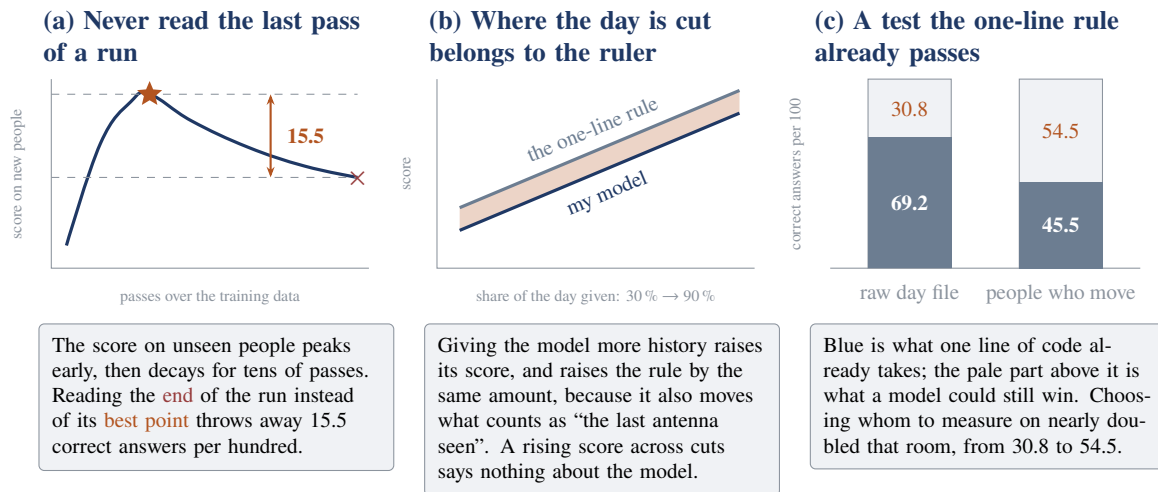


Figure 4.7: The three rules about measuring, and the worth of each

run, deliberately allowed to continue to a hard ceiling of forty-nine passes. Its training score improves the whole way through, but its score on people it had never seen climbs for nine passes, peaks, and then declines for the remaining forty. The copy saved at the end scored 48.7 % on the held-out people; the copy saved at the peak scored 64.2 %. Reading the end of training instead of its best point therefore throws away **15.5 correct answers per hundred**, and spends forty passes of waiting time doing so.

Acting on that, keeping the best copy, stopping on a plateau, and adding three mild settings that make memorisation harder,¹⁰ moved the score on unseen people from 48.7 % to **64.4 %**, cut the waiting time from seventy-two minutes to twenty-nine, and lifted top-3 from 76.2 % to 82.7 %. Panels (b), (c) and (d) confirm that this shape belongs to the task rather than to one run. So every run in this report is stopped early and read at its best copy, and two runs are always compared at their respective peaks.

¹⁰Three standard knobs. **Dropout**: a tenth of the added numbers are set to zero at random at each step, so the model cannot lean on any single one. **Learning rate**: how big a step to take when adjusting a number. **Weight decay**: a mild pull of every trainable number back towards zero.

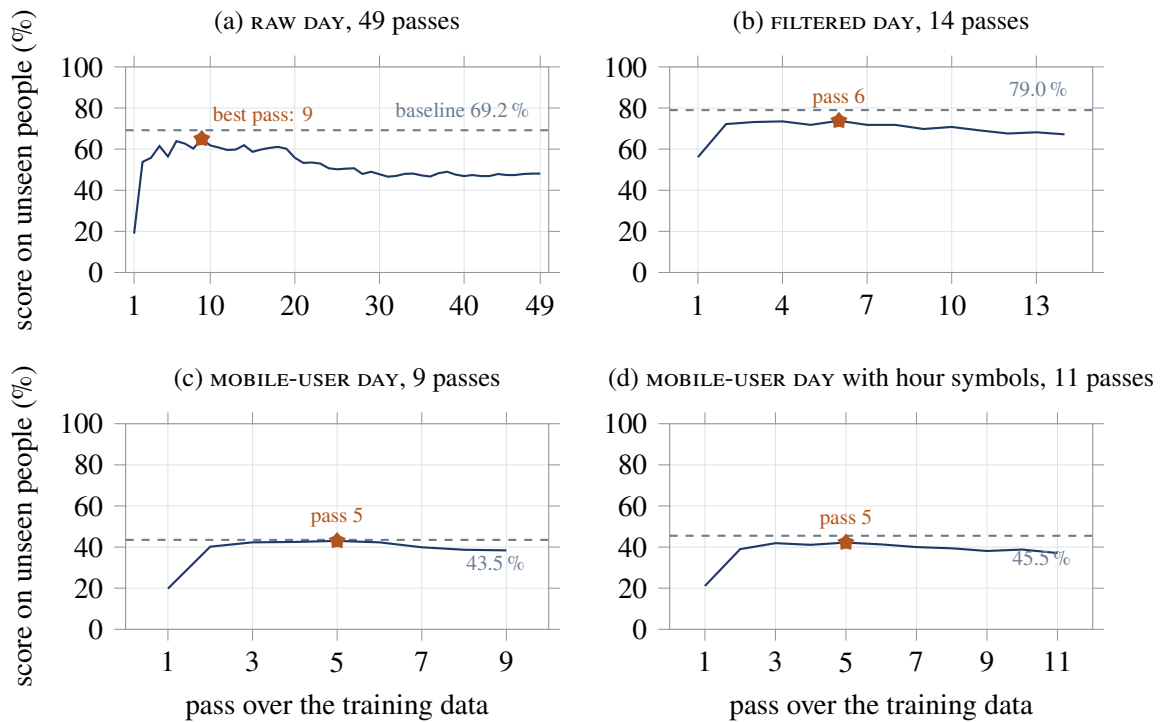


Figure 4.8: The score on unseen people, pass by pass, on four sets of people

What is plotted. Four separate training runs on four different sets of people. The horizontal axis counts the passes a run made over its training data; the vertical axis is the score on people it is *not* training on. The solid line is the model, the star marks its best pass, and the dashed line is the one-line rule on that same set of people. In all four the score climbs for four to nine passes, peaks, then declines, while the score on the run’s own training data keeps improving: that divergence is the model switching from learning how the population moves to learning its examples by heart.

What it costs to read the wrong pass, and a warning about the dashed line. In panel (a) the best pass scores 65.0 % and the last one 48.1 %, so reading the end of a run instead of its best point throws away 16.9 correct answers per hundred plus forty passes of waiting time; measured on the held-out people rather than the validation ones the same choice costs 15.5. The dashed line moves between panels, from 69.2 % on the unfiltered day to 43.5 % once only mobile people are kept, because the people are different. The four panels are comparable only on the distance between solid and dashed, and in all four that distance is negative: none of these runs beats the rule.

→ Produced by `train_cellid_llm.ipynb`, the training loop these per-pass scores come from.

4.5.2 Rule 2: the cut of the day belongs to the ruler

The share of the day given to the model, called the *context fraction*,¹¹ looks like a setting of the experiment. It is part of the measurement. Moving the cut changes which antenna counts as “the last one seen”, so it moves the one-line rule too. A model graded at 90 % context is not a better model than the same model graded at 30 %; it is the same model measured against an easier ruler.

Table 4.1 contains the mistake this rule exists to prevent. On the FILTERED DAY the model reaches 80.1 % at the reference 80/20 split against a rule at 79.0 %: **1.1 more correct answers**

¹¹**Context fraction:** the share of a person’s day handed to the model before it must produce the rest.

Table 4.1: Moving the cut moves the ruler with it

past / rest	Valid.	Model	Rule	Distance	top-3	top-5	Beats it?
<i>FILTERED DAY</i> , three separate trainings, one per cut							
70 / 30	72.3 %	76.5 %	76.8 %	−0.3	87.1 %	90.1 %	no
80 / 20	73.8 %	80.1 %	79.0 %	+1.1	89.1 %	91.6 %	yes
90 / 10	75.7 %	77.7 %	80.1 %	−2.4	87.2 %	90.5 %	no
<i>MOBILE-USER DAY</i> , one training at 80/20, evaluated at seven cuts							
30 / 70	43.0 %	41.5 %	42.9 %	−0.9	62.1 %	69.9 %	no
40 / 60	43.0 %	42.0 %	43.2 %	−0.6	62.1 %	69.7 %	no
50 / 50	43.0 %	42.0 %	43.3 %	−0.7	62.0 %	69.4 %	no
60 / 40	43.0 %	42.7 %	44.0 %	−0.9	62.1 %	69.5 %	no
70 / 30	43.0 %	43.6 %	44.3 %	−0.3	62.1 %	69.5 %	no
80 / 20	43.0 %	44.6 %	45.5 %	−0.9	63.9 %	70.7 %	no
90 / 10	43.0 %	47.6 %	50.9 %	−2.1	68.0 %	74.3 %	no

What was varied, and what the columns are. Only where each person’s day is cut into the part the model reads and the part it must produce: “70 / 30” means the model sees the first 70 % and is graded on the last 30 %. **Valid.** is the score on people held back to decide when to stop training, **Model** the score on the test people, **Rule** what “repeat the last antenna” scores on those very same moments, and **Distance** the model minus the rule, which is the only column that means anything.

The point of the table is the Rule column. It changes from row to row, because moving the cut changes which antenna counts as “the last one seen”. On *FILTERED DAY* exactly one cut out of three clears the rule, by 1.1, while the other two lose ground by 0.3 and 2.4; one favourable cell out of three is not a result, and the next iteration made it disappear. On *MOBILE-USER DAY* no cut clears the rule at all: the model climbs steadily from 41.5 to 47.6 % as it is given more history, which looks like progress until one notices the rule climbing just as fast. Each *FILTERED DAY* row is a separate training run; the *MOBILE-USER DAY* block is one model read seven times, and the giveaway is its identical **Valid.** column.

↔ Produced by `train_cellid_llm.ipynb`, the context-fraction sweeps.

per hundred, and the first time in the project that a trained model cleared the one-line rule. Read on its own that looks like a result. Read across the three rows it is far weaker: at 70 % context the model trails by 0.3, and at 90 % context, where its own score is the *highest* of the three, it trails by 2.4.

4.5.3 Rule 3: a test the one-line rule already passes measures nothing

On a day file taken as it comes, “repeat the last antenna” is right 69.2 % of the time, and most of what is left above that is not mobility at all: it is people with three records from a single antenna. A model scoring 64.4 % on such a file has not failed at prediction; it has been measured on a benchmark with no room in it. The same trained architecture, on three sets of people, faces rules of 69.2 %, 79.0 % and 45.5 %.

4.5.4 The three rules closed the tuning avenue

Taken together they closed the “tune it harder” avenue. Once runs were read at their peak, once the cut was recognised as part of the ruler, and once the people had been chosen so that the yardstick was beatable in principle, the model still sat within a few correct answers of a one-line rule, usually below it, at every cut and on every set of people tried.

The example that decided what to do next is small enough to quote in full. On one run, **person A**, whose past is the same antenna repeated ten times, is predicted correctly with 100 % confidence; **person B**, whose past ends on one antenna and whose next event is a different one entirely, is predicted as “the same antenna again” with 68 % confidence. The model was right for free on the first and wrong for free on the second. A companion experiment made the same point statistically: on a 100-person version of the data, the score per person ranged from 10 % to 100 % and the model never passed 42.1 %. One hundred people is not a population; it is a handful of anecdotes with an average printed underneath. That is what Chapter 5 sets out to fix.

4.6 Conclusion

This chapter fixed the instrument and then fixed the ruler. The instrument is a small model, Qwen2.5-0.5B, retrained with QLoRA on one free graphics card, fed a day written as one symbol per antenna and one symbol per hour, and graded only where an antenna name has to be produced. It was settled in the first week and never changed again, which is the only reason the eleven iterations can be compared with one another at all: what varied from one to the next was the data and the training signal, never the machine underneath.

The ruler cost more. Each of the three rules arrived *after* a conclusion that turned out to be wrong, and the 1.1 more correct answers per hundred that I presented as the project’s first result was not one. Reading a run at its best saved copy rather than at its last was worth 15.5 correct answers per hundred, far more than any modelling idea in this report; recognising that the cut of the day belongs to the measurement killed that apparent win; and choosing people on whom one line of code does not already score 69 % is what made the task measurable at all. A short checklist written before the first training would have caught all three at once, and the modelling work would have started two iterations earlier.

One thing I never went back to is the size of the part of the model that training is allowed to change. QLoRA only trains a small share of the numbers, about two in every hundred, and I left that share at the same value in every run, from four hundred training people up to twenty-one thousand. Fifty times more data was poured into a part that never grew, and Chapter 8 is where that shows.

5 The person sets the score, not the model

5.1 The goal

The same trained model, on the same run, predicts one person perfectly and another not at all. That makes an average score almost uninformative, and it replaces the project’s question with a sharper one: *what property of a person decides whether that person is predictable?* If the answer is a property of the model, there is a model to improve. If it is a property of the person, the average score is largely fixed before any training begins. This chapter establishes that it is the second.

Where the numbers in this chapter come from. Two sets of people, both in Table 3.1. The first is a day file taken as it comes, on which 1,173 people were scored. The second is the MOBILE-USER DAY, where 400 people were used for training and **100 different people**, never seen during training, for grading. Wherever a figure says “100 held-out people”, it means those hundred strangers.

5.2 The property of a person that decides the score

The candidates are all things computable from a person’s own day without any model: how often they change antenna, how much of their time is spent on one dominant antenna, how many different antennas they visit, how irregular their changes are, and how many records they have.

Each candidate was compared against that person’s own score using the **Pearson correlation**, written r . I did not invent it: it is the measure Karl Pearson published in 1895 [10] and it has been the standard one ever since. It is a single number saying how tightly two quantities move together across a population:

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x}) (y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (5.1)$$

Here n is the number of people; x_i is the candidate property measured on person i , for instance how often they change antenna; y_i is that same person’s score; and $\bar{x}$ and $\bar{y}$ are the averages of those two over everybody. The numerator is large when people above average on one quantity are also above average on the other, and the denominator rescales the result to land between -1 and $+1$. So $r = +1$ means the two rise together in lockstep, $r = -1$ that one falls exactly as the other rises, and $r = 0$ that they have nothing to do with each other.

Figure 5.1 ranks the five candidates by that coefficient, and one of them stands clear of the others: how often a person changes antenna. The winner needs a definition before it can be used. Take one person’s day and walk through their records in order, comparing each with the one just before. The **cell-switch rate** is the share of those comparisons in which the antenna is not the same. Somebody who spends the whole day at home has a rate near 0%,

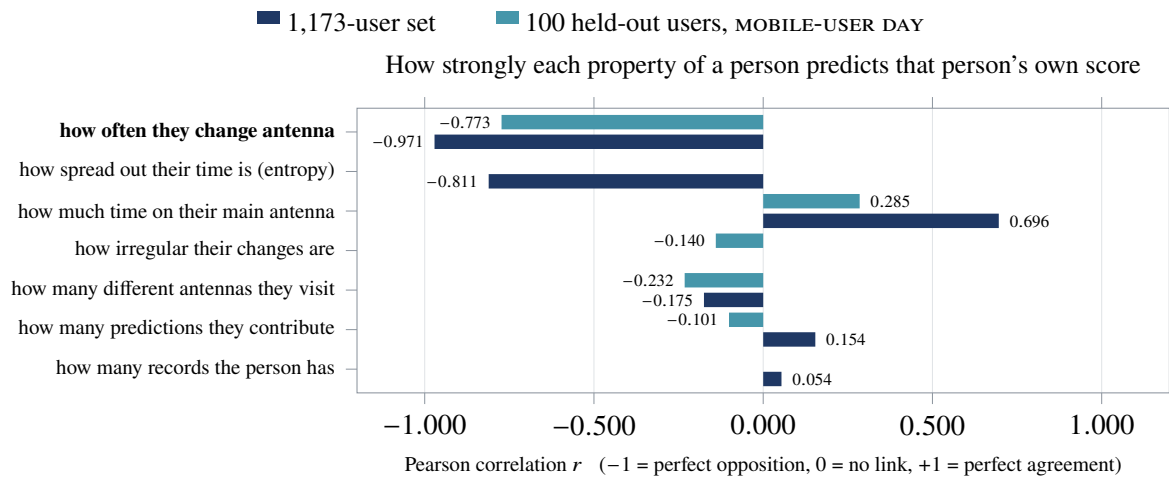


Figure 5.1: The one property that predicts a person's score

What each bar is. One row per candidate property of a person, named on the left in plain words. The bar is the Pearson correlation of Equation (5.1) between that property and that person's own score: zero means the property tells you nothing about the score, ± 1 that it tells you everything, and pointing left means “more of this, worse predicted”. Dark blue is measured on the 1,173 scored people of a raw day file, teal on the 100 held-out people of a different set.

The ranking is not close, and the top bar needs a warning. How often somebody changes antenna reaches -0.971 on the first set and -0.773 on the second; how spread out their time is (-0.811) and how much sits on one main antenna ($+0.696$) are the same information from other angles. Everything else, including how much history the person has, stays below 0.24. But -0.971 is close to arithmetic, since how often somebody changes antenna is the mirror image of how often the one-line rule is right on them, so any model imitating that rule earns a large negative value before anything is measured.

somebody constantly on the move near 100 %. It is a measure of restlessness, computed from the person's own day with no model involved, and it reaches $r = -0.971$ on the 1,173 people of the raw day and -0.773 on the 100 held-out people of the MOBILE-USER DAY: *the more often somebody changes antenna, the worse they are predicted.*

That -0.971 is the weakest number in this chapter, because it is very nearly arithmetic: the switch rate is by construction the mirror image of how often “repeat the last antenna” is correct on the same moments, and this particular model was behaving like that rule. What the coefficient really quantifies is how closely the run was imitating the thing it was supposed to beat, and nobody re-measured it once Chapter 7 finally cleared the rule.

What the figure does establish, and what no argument about tautology can take away, is the negative half of its ranking. The number of distinct antennas visited, the number of predictions contributed and the raw length of the day all stay below 0.24 in absolute value, so the intuitive explanations, that a hard person simply has less history or a wider repertoire of places, are eliminated in a single pass. Repetitiveness is the only candidate left standing.

5.2.1 The same finding, without any correlation

Figure 5.2 makes the point again without relying on r at all, simply by sorting people by their switch rate and cutting the sorted list into equal groups.¹

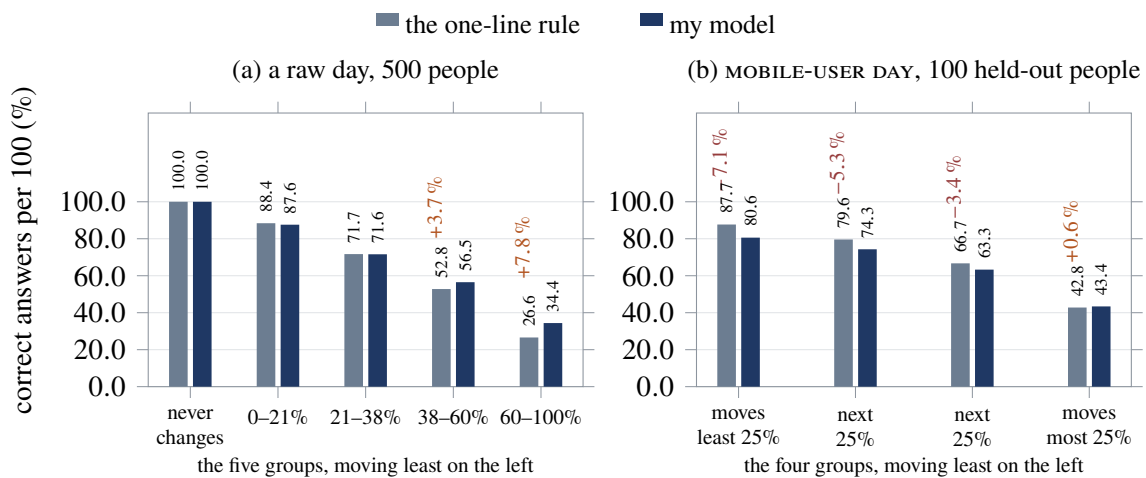


Figure 5.2: The score follows the rate at which a person changes antenna

How to read either panel. People are sorted by how often they change antenna, from those who move least to those who move most, and the sorted list is cut into equal groups, five on the left and four on the right; movement increases left to right in both. Grey-blue is the one-line rule, dark blue my model, both scored on exactly the same moments of the same people, and the number above each pair is the distance between them.

A spread of 65 in panel (a), and a less comfortable panel (b). The fifth of people who never change antenna are predicted at 100 % by both; the fifth that moves most plateaus at 34.4 %. That is 65 correct answers per hundred between the two ends of one population, on one trained model. The two bars are the same height in the three groups that move least: the model only pulls ahead in the two hardest, by 3.7 and 7.8, and those hold 18.5 % of the graded moments. On a second, independent set of people the model *loses* to the rule in the three quarters that move less, by 7.1, 5.3 and 3.4, and only draws level in the one that moves most. Same ordering, same mechanism, negative result overall.

One objection has to be closed before any of this can be called a property of the person rather than of the ruler: perhaps these people are only hard because of where their day happened to be cut. Table 6.1, in the next chapter, closes it. The same person's score with 30 % of their day in context and with 80 % of it correlates at $r = 0.77$, so difficulty travels with the person, not with the amount of history handed over.

5.2.2 The model helps in the restless minority, which is a small share

The same figure carries the second half of the finding, and it is the more useful half. In the three easiest groups the model and the rule are indistinguishable. It is only in the two hardest that the trained model pulls ahead, by 3.7 and then by 7.8 correct answers per hundred. The model's contribution is real, it is consistent, and it lives exactly where repetition stops working.

¹**Quintile:** one fifth of the population, obtained by sorting people on some quantity and cutting the sorted list into five equal parts. **Quartile** is the same with four.

It also lives exactly where the volume is not. Those two groups hold 18.5 % of all graded moments, and the most restless group alone, where the model gains 7.8, holds 5.6 %. Winning eight answers per hundred on one twentieth of the volume moves the overall average by less than half an answer per hundred. The consequence is a reporting rule rather than a modelling one, applied from this chapter onward: *an average score on this task is a weighted mixture of segments with wildly different difficulty, and quoting it alone hides the only part of the result that is about the model.*

5.3 A tenth of the people decide the average

If the score varies that much between people, then how much each person weighs in the average matters, and the weights turn out to be extremely uneven. Splitting the same 1,173 people into five equal groups by how many predictions each contributes, the middle person contributes eleven predictions while the busiest tenth, 117 people, carries **58.6 %** of the 54,173 graded moments. Any average on this data is, to a first approximation, the score of those 117 people.

The tempting conclusion, that having more records makes somebody more predictable, is not supported. The score is almost flat across the five volume groups, and the raw correlation between volume and score is only $r = +0.154$. What actually happens is that people with many records tend to change antenna *less* often, $r = -0.15$ between the two, so the mild upward tilt at the busy end is the switch-rate finding wearing a different label. It also means that levelling the number of records per person cannot be expected to redistribute difficulty, which is exactly the prediction Chapter 8 tests head on.

5.4 The target is the antenna name, not the mast

Looking at people one by one also exposes something an average cannot: part of the remaining error is not a modelling failure at all, but a deliberate property of the measurement.

Table 5.1 lists the seven worst-scoring people of that run, each with the reason it scores badly. The pair at the bottom of it is what mattered for the rest of the internship. Those two people oscillate between near-twin antennas of the *same* mast, so plain repetition scores 50 to 67 % on them while the model, which tries to choose between the two names, scores nothing at all.

The decision behind that looks like an oversight and is not one. *Movement*, for every behavioural quantity in this report, is measured on the mast, which is what makes the four kinds of user of Chapter 7 meaningful. The *prediction target* remains the full antenna name, because that is the question that was actually asked, and because coarsening it halfway through would have made every number measured before that point incomparable with every number after it. A small, known share of the reported error is therefore the model being marked wrong for a difference that is not a movement.

Table 5.1: The seven worst-scoring people, and the reason each scores badly

user_id	Moments	Ceiling	Rule	Model	Which situation
1049149	3	0 %	0 %	0 %	A, two to twelve moments each: too little material to judge a model
1291964	3	0 %	0 %	0 %	
1344347	8	25 %	12 %	12.5 %	
929903	12	75 %	8 %	8.3 %	
1161518	9	89 %	67 %	11.1 %	B, a real change of behaviour
828479	2	100 %	50 %	0 %	C, two names, one mast
1067194	3	100 %	67 %	0 %	

Who is in this table, and what the columns say. These are **the seven worst-scoring people of the whole test set** on FILTERED DAY, and nobody else: every person whose score is at or below 12.5 %. **user_id** is the anonymous number the operator assigned to that person for that day. **Moments** is how many graded moments they contribute. **Ceiling** is the best score reachable at all given the antennas that appear in that person’s own history: if one never appeared before, no model could have named it.

One bad number, three unrelated problems. Four of the seven contribute between two and twelve moments each, and a person with three moments can only score 0, 33, 67 or 100 %. One is a genuine miss: the ceiling is 89 %, the behaviour was learnable, and neither the rule nor the model followed it. The last two are the row that mattered. Their ceiling is 100 %, plain repetition already scores 50 to 67 %, and my model scores nothing at all, because they bounce between near-twin antennas of the *same* mast, UKVCHE102 and UKVCHE104 for instance, so it is marked down for failing a distinction that involves no movement.

↔ Produced by `train_cellid_llm.ipynb`, the per-user error listing.

5.5 Three statements survive, and they point to the time of day

Three statements survive this chapter. The score of a person is set by how repetitive their movement already is, and by nothing else I measured. The model’s real contribution is concentrated in the restless minority, which owns a small share of the graded volume, so the average can barely move even when the contribution is real. And a known share of the remaining error is the model being marked wrong for a distinction between two names of the same physical place.

Taken together they say that *the lever is not capacity*. A bigger model, or a longer training, would improve predictions in the segment that repetition already solves for free, which owns most of the volume and none of the difficulty. What was needed instead was a signal that helps specifically *at the moment a person is about to move*, and the most obvious candidate is the time of day.

5.6 Conclusion

The chapter set out to find whether the score belongs to the model or to the person, and the answer is unambiguous: to the person. Of everything computable from somebody’s own day,

only one quantity tracks how well they are predicted, and it is how often they already change antenna. The number of distinct places they visit, the length of their day and the number of predictions they contribute are all near zero, which disposes of the two intuitive explanations at once. The model's own contribution is real but narrow: it appears in the two most restless groups, worth 3.7 and then 7.8 correct answers per hundred, and those groups hold 18.5 % of the graded moments between them.

I led this chapter, at the time, with $r = -0.971$, and that was a mistake. A coefficient that is very nearly arithmetic, because the switch rate is the mirror image of how often the trivial rule is right, cannot carry a chapter however striking it looks; at best it measures how closely that particular run was imitating the rule it was meant to beat. What genuinely changed what I did next is the *distribution*: one fifth of the people predicted perfectly by doing nothing at all, one fifth plateauing at 34.4 %, and an average written by 117 people out of 1,173. The correlation was the headline; the histogram was the finding.

The practical consequence is the reporting rule this report has followed ever since. An average score on this task is a mixture of segments of wildly different difficulty, so it is quoted here only alongside the one-line rule measured on the same people, and the interesting half of any result is which segment the gain came from.

6 Adding the time of day

6.1 The goal

Up to this point the model was given a bare list of antennas. Nothing in its input said whether an event happened at four in the morning or at seven in the evening, which is a strange omission for a mobility problem: the reason a person changes antenna at all is usually that they are going somewhere, and going somewhere happens at particular times of day. So the question of this chapter is whether *time of day* is the missing signal, and whether a model can be pushed to expect a change of antenna at the hours where a change is likely, instead of falling back on the one strategy always available to it.

Where the numbers in this chapter come from. All of them from the MOBILE-USER DAY (Table 3.1): 400 people used for training and **100 different people**, never seen during training, used for grading. The one-line rule scores 45.5 % on the last fifth of their day and 43.1 % when every position of the day is graded, and the second half of this chapter switches to the second of those two protocols.

6.1.1 The clue: some people collapse at the end of the day

The pointer came from looking at people one at a time. Scoring the same trained model with progressively more history, most behave exactly as Chapter 5 predicts: their score is a trait, roughly constant whatever they are given. But one group does something else entirely. Given *more* history, their score falls off a cliff, from thirty or forty per hundred down to zero.

Table 6.1 puts eight of those people side by side, and its bottom four are the ones worth looking at: given most of their day rather than a third of it, they score zero or nearly. So the missing ingredient is not more history. It is knowing that the end of the day is not the middle of it.

Table 6.1: Eight people, and the four whose score collapses when given *more* history

user_id	Short past	Long past	Change	Which situation
2046755	12.1 %	10.8 %	−1.3	A, difficulty is a trait of the person: more history changes nothing
599692	16.5 %	16.7 %	+0.2	
1243093	8.3 %	15.6 %	+7.3	
1367716	60.4 %	37.5 %	−22.9	B, a real decline, still usable
401222	32.2 %	0.0 %	−32.2	C, total collapse at the end of the day: the model repeats the last antenna with up to 78 % confidence
1300439	40.7 %	0.0 %	−40.7	
1223357	33.7 %	4.2 %	−29.6	
1153529	20.0 %	3.0 %	−17.1	

What was done, and what should have happened. One trained model, one set of people, scored twice: once with a short slice of each person’s day as history, 30 to 60 % of it, and once with a long one, 80 to 90 %. The **Change** column is the second score minus the first. More history should help, or at worst do nothing, and that is what the top three rows show. Across all people the two scores agree at $r = 0.77$, which is what makes difficulty a trait of the person rather than an effect of where the day was cut.

What happens to the bottom four, and why it mattered. Their score does not merely decline, it falls to zero or nearly, and *which* moments a long history is scored on is the clue: with 80 to 90 % of the day given, the moments left to answer are the ones at the *end* of it. A model that follows somebody correctly through the morning and then scores nothing on their evening is not short of capacity. It is confidently repeating the last antenna it saw, with up to 78 % confidence in several of these cases, at exactly the moment that person stops staying put. MOBILE-USER DAY, one trained model evaluated at several cuts.

↪ Produced by `train_cellid_llm.ipynb`, the per-user, per-context evaluation pass.

6.2 The form the time of day takes, and the two bets it carries

Time enters in the form fixed in Chapter 4: one symbol for the hour placed before each event, always given to the model and never predicted by it. Two distinct bets were made at once, and only one of the two survived the chapter.

Bet 1: the model can use the hour. Given the hour of every past event and of the event to be named, the model can learn that some antennas are morning antennas and others evening ones.

Bet 2: some hours deserve more weight. There are *transition windows*,¹ stretches of the day in which a person is more likely to move between the area they live in and the area they spend the day in. If mistakes there are made to count more, the training is forced to get them right rather than accepting them as the price of a good average.

Bet 2 requires knowing which hours those are, and at this stage I did not measure them. I *guessed* them, from the time boundaries the team’s home-and-activity classification already used: **04–06h** and **18–20h**. That guess is the subject of the end of this chapter.

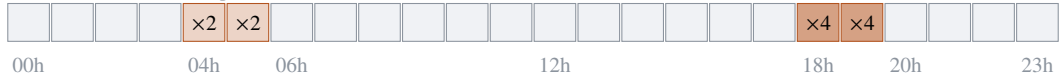
¹**Transition window:** a stretch of hours in which a mistake is made to count more, so that a wrong answer at 19h under a multiplier of four costs four times a wrong answer at 14h.

6.2.1 Bet 1 holds: the model does read the hour

Whether a model uses an input is not something to assume. It is something to test by breaking the input and seeing whether the score notices.

(a) The 24 hours of the day, and how much a mistake costs in each

A plain square means an ordinary mistake. A shaded one means a mistake there counts two or four times as much. No data is duplicated.



(b) What was done to the hours, and only to the hours

The same trained model, the same test people, the same sequence of antennas. Only the hour symbols change.

hours left alone	<H07>	<H09>	<H12>	<H18>	the reference
hours shuffled	<H12>	<H18>	<H07>	<H09>	real hours, wrong order
all hours set to 12h	<H12>	<H12>	<H12>	<H12>	no information at all
every hour moved by +12h	<H19>	<H21>	<H00>	<H06>	order kept, labels wrong

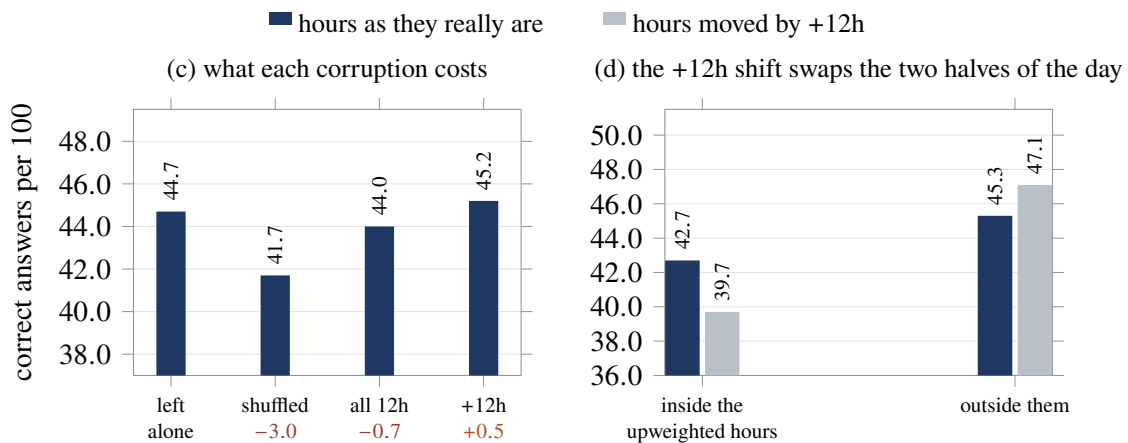


Figure 6.1: The model does read the hour

Panels (a) and (b): breaking the hour, and only the hour. Each event enters the sequence as an hour symbol followed by an antenna name, and mistakes in two chosen stretches of the day were made to count more, twice as much at 04h and 05h and four times as much at 18h and 19h. Panel (b) re-grades the same trained model on the same test people three more times, never touching the list of antennas: the hours are shuffled, so they are real hours attached to the wrong events; or all replaced by one value, which deletes the information; or all moved forward by twelve, so the ordering survives but every label is wrong.

Panels (c) and (d): the argument is the gap between two bars. Deleting the hour costs 0.7 correct answers per hundred; merely shuffling it costs 3.0. If the hour symbol were just occupying a slot, destroying its *ordering* could not cost more than deleting it altogether, so the model has genuinely learned which hour goes with which antenna. Moving every hour forward by twelve changes the overall score by only +0.5, which is not indifference: inside the two upweighted stretches the score falls from 42.7 to 39.7 while outside them it rises from 45.3 to 47.1. The two halves of the day exchanged their difficulty along with their labels. MOBILE-USER DAY, 80 % of each day given as history, 1,747 graded moments.

The test is the one drawn in Figure 6.1: the hour symbols are deliberately corrupted in three different ways, and the score is re-measured each time. The argument is the difference between two of those numbers, and only that. Replacing every hour by a constant, which removes the information completely, costs 0.7 correct answers per hundred. Merely shuffling the hours, so that the values are still real hours but attached to the wrong events, costs 3.0, four times as much. If the hour symbol were simply occupying a slot in the sequence, destroying its *ordering* could not possibly cost more than deleting it altogether. Bet 1 is established.

6.2.2 Bet 2 half-holds: the hours can be won, at a price

Three changes were made together. **An uneven weight:** the two windows are not comparable in size, since 18–20h carries four times as many records as 04–06h and touches 58 % of people against 39 %, so 18–20h was raised to a multiplier of four while 04–06h stayed at two. **Ask the question more often:** training slices containing at least one in-window move were drawn three times more often than the rest,² which raised the share of training batches containing such a move from 45.1 % to 71.2 % without duplicating a single record. And **grade the whole day:** until then only the last fifth of each test person’s day was graded, and from this point every position is, one at a time. That is strictly harder, because early positions have almost no history behind them, and it is why the one-line rule on these same people is quoted as 43.1 % on the full day against 45.5 % on the last fifth.

Figure 6.2 and Table 6.2 measure what those three changes were worth: the first group of moments by group of moments, the second at seven different cuts of the day.

Table 6.2: The same change, measured everywhere else: seven cuts, seven losses

Past given	Distance before	after	Change	top-3, before → after	top-5, before → after
30 %	−0.9	−1.6	−0.7	61.4 % → 58.8 %	69.1 % → 65.7 %
40 %	−0.6	−1.5	−0.9	61.7 % → 58.8 %	69.0 % → 65.6 %
50 %	−0.7	−1.1	−0.4	61.4 % → 58.8 %	68.6 % → 65.4 %
60 %	−0.9	−1.5	−0.6	61.7 % → 58.7 %	68.7 % → 65.3 %
70 %	−0.3	−1.3	−1.0	61.4 % → 58.5 %	68.6 % → 65.2 %
80 % (<i>ref.</i>)	−0.9	−1.3	−0.4	62.3 % → 59.1 %	70.2 % → 65.4 %
90 %	−2.1	−2.8	−0.7	64.6 % → 60.6 %	72.3 % → 67.7 %

What is compared. The same 100 held-out people and the same grading, before and after the three changes described above. Each row is one cut: “30 %” means the model was given the first 30 % of each person’s day as history. These are the moments *outside* the two emphasised stretches, the great majority of them. **Distance before** and **after** are the model minus the one-line rule, in correct answers per hundred.

Why seven rows and not one. A single degraded measurement would be noise. Seven, at seven different cuts, all moving the same way, is a model that has been narrowed: better at the hours it was told to care about and worse everywhere else. Top-3 and top-5 fall by between 2.6 and 4.8 across the board, so this is not the right answer slipping from first to second place. It has left the shortlist entirely.

↪ Produced by `train_cellid_llm.ipynb`, the targeted-sampling run, evaluated at seven cuts.

²**Sampler:** the component that chooses which slice of training data to show next. Making it *weighted* means some slices are drawn more often, which changes how often a question is asked rather than how much a wrong answer costs.

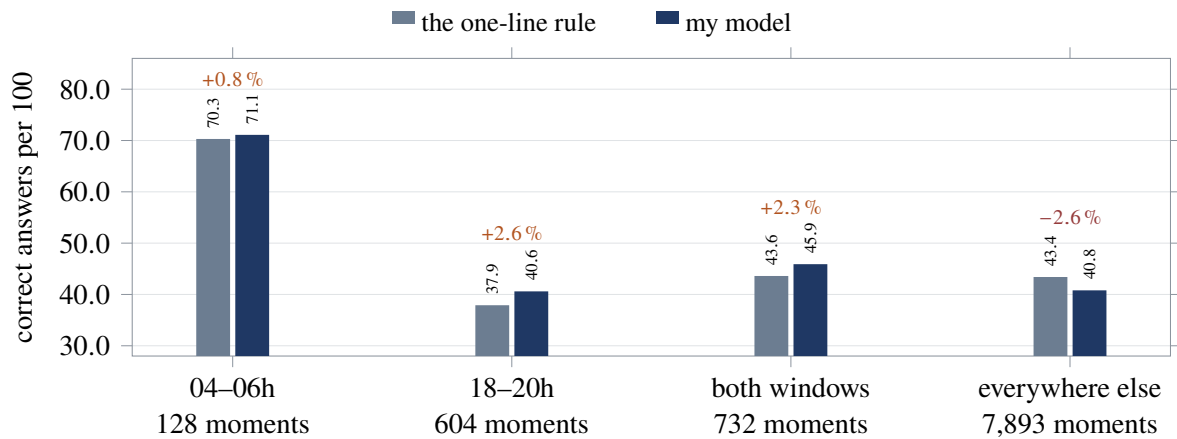


Figure 6.2: Emphasising two stretches of the day: the win, and the price

How to read the four pairs. Each pair is one group of graded moments, with the number of moments printed under its label so the weight it carries is visible. Grey-blue is the one-line rule, dark blue my model, on exactly the same moments. Before this run, training slices containing a move inside one of the two chosen stretches of the day were shown three times more often than the rest, and a mistake at 18–20h was made to cost four times an ordinary one.

The win is real; the price is larger. At 18–20h the model gains 2.6 correct answers per hundred, and 2.3 over the two chosen stretches combined, the first time in the project that a trained model beat “repeat the last antenna” on a segment chosen *in advance*. But the fourth pair is everything outside those stretches: 7,893 moments, more than ten times the volume of the win, and the model loses 2.6 there. The trade is only worth making if the emphasised hours really are the hard ones, and Table 7.1 shows they were not. MOBILE-USER DAY, 100 held-out people, every position graded; the 04–06h bar rests on 128 moments only and should be read as “not worse” rather than as a win.

6.2.3 Right mechanism, wrong hours

As Figure 6.2 shows, the model gains 2.6 correct answers per hundred inside the 18–20h stretch and loses ground almost everywhere else, and the seven rows of Table 6.2 all move the same way. The gain and the cost are not contradictory. Weighting a chosen set of moments does exactly what it is designed to do: it buys correct answers there and pays for them elsewhere. The mechanism was never in question after the ablation. What was in question, and what I had never checked, was whether the moments I had chosen were the hard ones.

They were not, and the measurement is unambiguous. On the full test day the one-line rule scores 43.6% *inside* the guessed windows and 43.4% *outside* them, a difference of two tenths. The global window 04–06h / 18–20h does not separate easy moments from hard moments at all, so every extra unit of effort spent there was spent on moments no harder than average. Worse, broken down by kind of user, 04–06h turns out to be *easier* than average for every single kind: the rule scores between 49 and 64% inside it, against 20 to 57% on average. Song et al. had published exactly that in 2010, their curve of hourly regularity being at its highest during the night and dipping around midday and in the early evening [13]. The window I guessed is the one stretch of the day the literature already marked as easy.

The machinery worked, and I pointed it at the wrong hours. The 2.6 won at 18–20h was real, because 18–20h happens to be genuinely difficult for one of the four kinds of user;

the loss on everything else was the cost of upweighting hours that were difficult for nobody. That has an obvious consequence, and it is what Chapter 7 does. If the right hours differ from one person to the next, the remedy is not to pick better hours for everybody. It is to stop assuming everybody shares the same hours.

6.3 Conclusion

The chapter carried two bets and settled both. The first holds: the model does read the hour, and the proof is that scrambling the hours costs four times as much as deleting them, which could not happen if the symbol were merely occupying a slot. That result is kept, and every run from this point on is given the hour of day. The second bet half-holds. Making mistakes count more inside a chosen stretch of the day does exactly what it is designed to do, buying 2.6 correct answers per hundred at 18–20h, but it pays for them everywhere else, at all seven cuts of the day tested. The machinery was right and I pointed it at the wrong hours: on these people the one-line rule scores 43.6 % inside the guessed windows against 43.4 % outside them, so the moments I paid to emphasise were no harder than average. Song et al. had published the hourly curve that says so in 2010, and the stretch I guessed is the one the literature already marks as easy.

What I take from that is a habit rather than a result. The hours were the only component of the design I never measured before relying on it, and they are the only one that failed; measuring them would have cost an afternoon against the iteration it cost instead. There is also a methodological cost I would avoid a second time. Grading every position instead of only the last fifth was the right choice, since leaving the early positions out was quietly making the task easier than the question deserved, but it moved the one-line rule on the very same people from 45.5 % to 43.1 %, so runs before and after cannot be compared on absolute score at all. A change of measurement is a change of ruler: I would freeze the grading protocol before the first training and treat any later change as opening a new series of results.

The failure is also what produced the rest of the report. If one pair of hours applied to everybody marks nothing, the remedy is not a better pair of hours. It is to stop assuming that everybody shares the same ones.

7 Sorting people into four kinds of user

7.1 The goal

Chapter 6 ended on a specific failure: one pair of stretches of the day, guessed in advance and applied to everybody, marks moments that are no harder than average. The remedy is not “pick better hours” but “stop assuming everybody has the same hours”. Some people leave home at seven in the morning and are settled by six in the evening; others do the opposite; a third kind barely moves at all.

That turns into two questions. *Can people be sorted into kinds of user according to when they move, using only information that will still be available at the moment a prediction is asked for?* That is a measurement problem, and the answer is yes. *And does telling the model which kind of user it is looking at make it predict that person better?* That is a modelling problem, and the answer was no until the kind of user stopped being a piece of text.

Where the numbers in this chapter come from. From the MOBILE-USER DAY (Table 3.1): the four kinds of user are fitted on the **400 training people**, and every score is measured on the **100 held-out people**, who are different people and never trained on. Every position of their day is graded, and on that protocol the one-line rule scores 43.1 %.

7.2 A kind of user is a shape, and it must be readable from the past

A kind of user here is not a label anybody assigned. It is a *shape*. For each person I compute how often their *physical location* changes, separately over four stretches of the day: night from 22h to 07h, morning from 07h to 12h, day from 12h to 18h, and evening from 18h to 22h. Those four numbers are the person’s profile. Four stretches rather than eight because each one has to hold enough events to be estimated at all.

Two properties of this definition matter more than the sorting itself. **Movement is measured on the mast, never on the antenna name**, because 12.4 out of every 100 back-to-back records change antenna name with no change of place (Chapter 3); measuring on the raw name was tried first and separated behaviour less well. And **the kind of user must be recoverable from the past alone**, since at the moment a prediction is asked for only the person’s history exists. The profile is therefore computed on the history only, never on the whole day, both during training and at grading time. Without that the idea would be circular.

7.3 The four kinds of user, and the hours each one finds hard

The sorting is done by an automatic procedure¹ handed each person’s four numbers and asked for four average profiles. **The number four was chosen rather than discovered**, and it is

¹**Clustering:** an automatic procedure that sorts items into groups by resemblance, with nobody labelling anything. Each group is summarised by its average profile, and each item is assigned to the nearest one.

the one arbitrary parameter of the whole mechanism: I picked it because four stretches of the day give four numbers per person, and because fewer than four profiles merged the two mirror-image kinds into one.

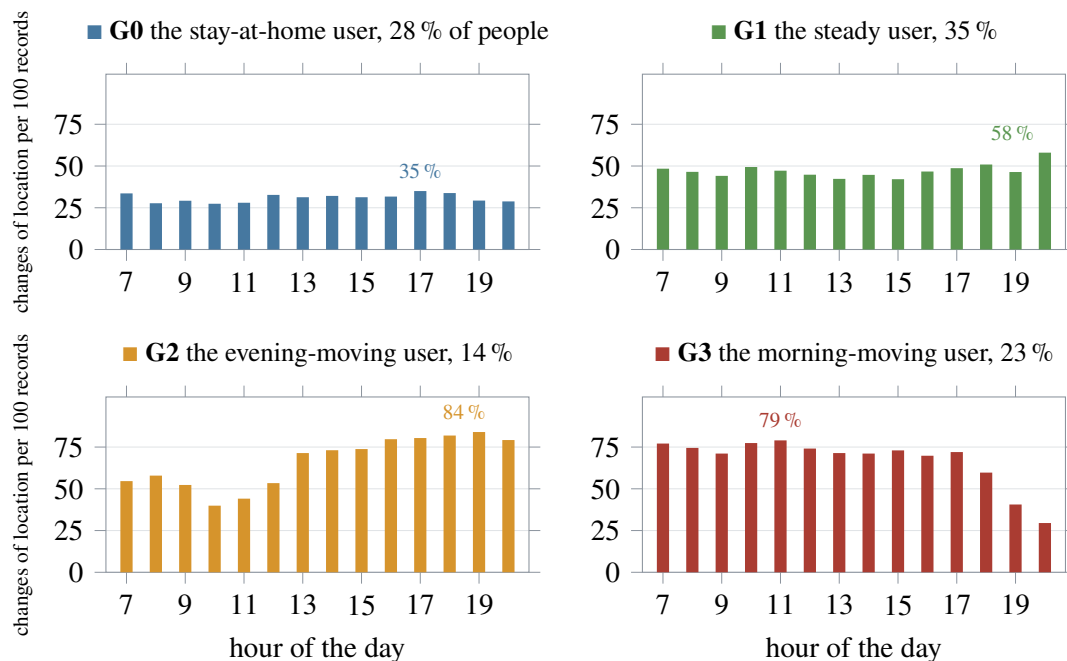


Figure 7.1: The four kinds of user, hour by hour

What each bar is. The horizontal axis is the hour of the day. Each bar counts, out of every 100 back-to-back records written during that hour, how many landed on a *different mast* from the one before: how often, at that hour, these people actually went somewhere. The percentage in each title is the share of *people* in that kind, not of records. Nobody labelled the four kinds; an automatic procedure was handed each person's rate over four broad stretches of the day and asked for four average shapes, and the names are mine.

Why G2 and G3 are the whole point. G0 is flat and low, around 30 changes per 100 records; G1 flat and higher, about 45 all day. G2 climbs all afternoon to 84 at seven in the evening; G3 does the exact opposite, 79 before noon then down to 29. They are near mirror images, they hold 37 % of the population between them, and *any hour that is hard for one is easy for the other*. MOBILE-USER DAY, 400 training people; the four labelled peaks are the values printed on the source chart and the remaining bars are recovered from its shape.

Figure 7.1 draws the four average profiles hour by hour, and the four shapes are easy to tell apart by eye.

G0, the stay-at-home user, 28 % of people. Flat and low all day, about 30 changes of location per 100 back-to-back records at every hour alike.

G1, the steady user, 35 %. Also flat, but higher: about 45 changes per 100 records from morning to evening. What characterises this person is the flatness, not the level.

G2, the evening-moving user, 14 %. A calm morning, then a climb all afternoon to 84 changes per 100 records at seven in the evening. Somebody who goes home late, or goes out after work.

G3, the morning-moving user, 23 %. The mirror image: 79 before noon, settling to 29 by evening. Somebody who commutes early and then stays put.

G2 and G3 are the reason a single global stretch of the day could never have worked. They are near mirror images in time, they are 37 % of the population between them, and any hour that is hard for one is easy for the other.

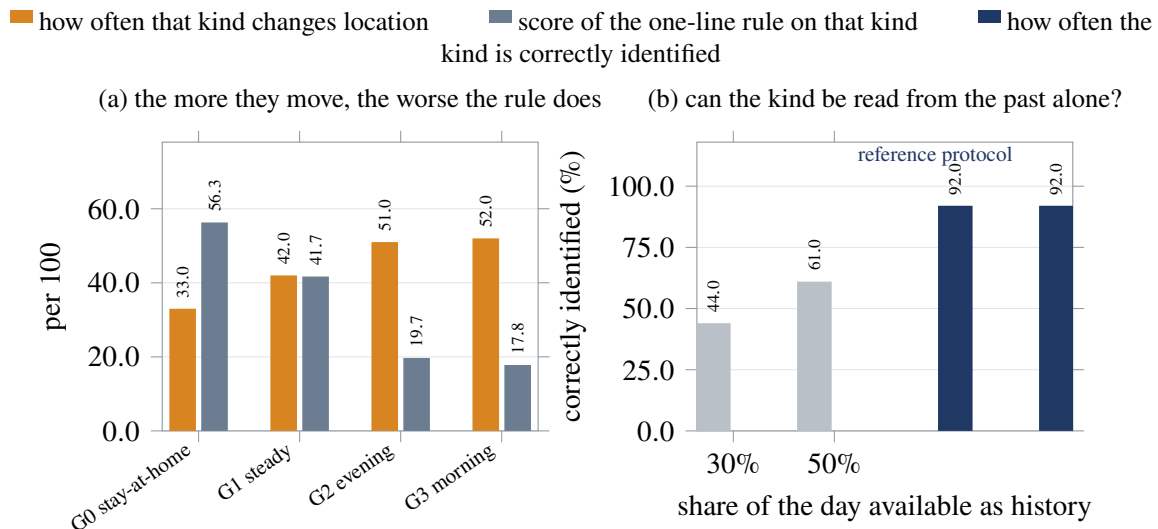


Figure 7.2: The kind of user predicts difficulty, and can be read from the past

Panel (a): the label says in advance how hard somebody will be. Two bars per kind of user, on two different scales, so read each colour against its own neighbours. Orange is how often that kind changes location over the day; grey-blue is what the one-line rule scores on them. The two rankings are exact opposites, with no exception: the stay-at-home user changes location about a third of the time and the rule is already right 56.3 % of the time for them, while the morning-moving user changes half again as often and the rule is right only 17.8 %.

Panel (b): the constraint that makes the idea usable, and where it breaks. At the moment a prediction is asked for, only the person's past exists, so the kind of user has to be worked out from that past or the exercise is circular. These bars measure how often the kind read from the past matches the kind computed from the complete day. Given 80 % of the day it is right for 92 of the 100 held-out people; below roughly 60 % it collapses, 61 % at half a day and 44 % at 30 %. That is worth nothing on somebody who has just appeared, which is exactly when an operator would most want a prediction. MOBILE-USER DAY, 400 training and 100 held-out people.

↪ Produced by `train_cellid_llm.ipynb`, the group-detection and per-group baseline cells.

Figure 7.2 carries the two results that make the sorting usable at all: its panel (a) shows that the label predicts *difficulty*, and its panel (b) that the label can be read from a person's past rather than from their whole day. Take the first. Knowing nothing whatever about a person except which of the four kinds they fall into accounts for 69 % of the spread in how predictable people are on the training side, and 72 % on the held-out side, with the profiles fitted on the training side only. That scale runs from 0 %, where the label would tell you no more than always guessing the population average, to 100 %, where it would tell you each person's score exactly.

One rival explanation has to be acknowledged: the sheer length of somebody's day also predicts how well they are scored, and length alone explains 48 % of that same spread.² But

² $\log n$: the logarithm of a person's number of records. Using it rather than n says that what matters is the

the label still explains 38 % of what remains after length has been removed, and the part of the profile that has no scale in it, how lopsided it is between morning and evening, is barely related to length at all.

7.3.1 Hours derived from the data, not guessed

With profiles in hand, the stretches of the day to emphasise can be defined the way they should have been from the start: as the run of consecutive hours where a given kind of user's *own* rule score falls furthest below that same kind's *own* average. In plain terms, the hours at which “repeat the last antenna” stops working *for that kind of person*.

Table 7.1: Hours derived per kind of user, and the hard moments they land on

Kind of user	Hours derived from the data (cost multiplier)	What it says
G0 stay-at-home user	<i>none</i>	a flat day has no hours worth emphasising
G1 steady user	18–19h $\times 1.4$, 20–21h $\times 1.7$	mild, and late in the day
G2 evening-moving user	13–14h $\times 1.7$, 16–21h $\times 2.5$	the five-hour block was later narrowed to 18–21h
G3 morning-moving user	07–08h $\times 2.0$, 10–13h $\times 2.3$	the mirror image of G2, and almost disjoint from it
Do those hours actually mark the hard moments? Score of the one-line rule, inside versus outside		
The guess of Ch. 6, 04–06h / 18–20h		43.6 inside vs 43.4 outside: a difference of 0.2 , so it marks nothing
The hours derived per kind of user		19.0 inside vs 46.4 outside: a difference of 27.4 , the genuinely hard hours

Where these hours come from. They are not chosen. For each kind of user I take the score of the one-line rule hour by hour *on that kind alone*, and keep the run of consecutive hours where it falls furthest below that kind's own average: the hours at which “repeat the last antenna” stops working *for that sort of person*. The multiplier says how much more a mistake costs there. The first row is what a hand-made rule could never produce, since the stay-at-home user gets no emphasised hours at all, and the last two rows are almost disjoint in time.

The lower block is the test that justifies the whole detour. An emphasised stretch is only useful if it lands on hard moments. The guess of Chapter 6 separates the rule by two tenths, which is to say not at all. The hours derived per kind of user separate it by 27.4, marking moments on which the trivial rule is right 19 % of the time against 46 % elsewhere. Worse, the guess was misplaced rather than merely imprecise: 04–06h is *easier* than average for all four kinds.

Table 7.1 gives the stretches that procedure returns for each kind of user, and its lower block is the measurement that justifies the whole detour. Applied to the full test day, the guessed global window separates the rule by two tenths: 43.6 % inside it against 43.4 % outside. The per-kind hours separate it by **27.4**: 19.0 % inside against 46.4 % outside. The

order of magnitude of a history.

first marks nothing at all; the second marks moments that are twenty-seven correct answers per hundred harder than average.

7.4 Passing the kind of user to the model

7.4.1 The first attempt failed, and the check that located the failure

Given all that, the first attempt was expected to work and did not. The kind of user was written into the input as a dedicated symbol placed at the head of the sequence, one shared symbol $\langle G0 \rangle$ to $\langle G3 \rangle$, so that the model could look back at it from every later position.³ The run scored 41.3 % against a rule of 43.1 %, so 1.8 behind, no better than anything before it.

The useful part is the check that followed, because it does not merely say “it did not work” but says *where*. Switching the symbol off entirely gives 41.1 %. Feeding a deliberately *wrong* kind of user gives 41.2 %. The correct one gives 41.3 %. Three conditions within two tenths of each other means one thing only: **the model was not reading the symbol at all**.

So the failure was not in the sorting, which had just been shown to separate the rule by twenty-seven, and not in the hours, which an earlier test had shown the model does use. It was in the *channel*: one symbol at the head of a sequence of up to five hundred is one symbol among many, and the model is free to dilute it to nothing, because ignoring it cost almost nothing during training.

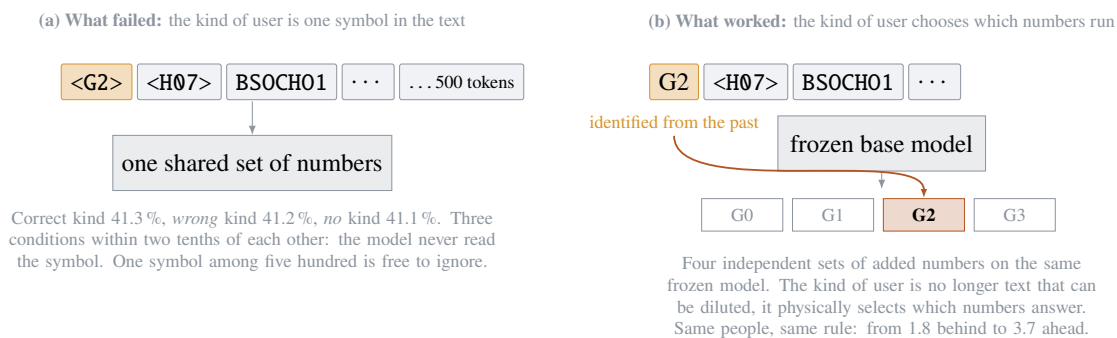


Figure 7.3: The kind of user as a piece of text, and as a switch

Panel (a): as a piece of text, and it was ignored. Same people, same protocol, same emphasised hours in both panels; only the channel differs. In the first attempt the kind of user was one extra symbol, $\langle G0 \rangle$ to $\langle G3 \rangle$, at the head of a sequence of up to five hundred. Feeding the correct kind, a deliberately wrong kind, or none at all gave 41.3, 41.2 and 41.1 %: three numbers inside two tenths, so the model was not reading it.

Panel (b): as a switch, and what it was worth. Four separate sets of added numbers are trained on the same frozen model, one per kind, and the kind read from a person’s past decides which set runs. There is nothing left to dilute, and the distance to the one-line rule moved from 1.8 behind to 3.7 ahead. Four sets are also four times the trainable capacity of one, and the control that would separate capacity from sorting was not run (Section 7.5).

Figure 7.3 draws the difference between the two ways of telling the model which kind of user it is looking at: on the left the kind of user as a symbol in the input, which the model

³**Attention:** the mechanism by which each position of the model looks at the others and decides how much weight to give each. It may only look backwards, never at the future.

may ignore, and on the right as a switch that decides which trained numbers run at all.

7.4.2 The fix, and the one part of it that mattered

Four changes were made. Three are bookkeeping, and one changes the nature of the mechanism.

1. **Narrow the evening-moving user's stretch** from 16–21h to 18–21h, since most of a five-hour block carries no more movement than the rest of the day.
2. **Work out the peak hours person by person, then average**, instead of from one histogram over everybody: because a handful of people own a large share of the records, the global histogram was really *their* day rather than the population's.
3. **Give each kind of user its own set of trained numbers.** The base model stays frozen and four independent sets of added numbers are trained, one per kind of user. The kind of user is no longer text the model may or may not attend to: it physically decides which numbers run. Each set is trained on every person of its kind and on nobody else, which turns four hundred people into four cross-trained populations⁴ rather than four hundred independent fits. Somebody with sixty events now benefits from everybody whose day has the same shape.
4. **Keep the derived hours.** The one component already measured to be correct was left untouched.

Only the third could plausibly account for a large move, and that is what the result suggests happened.

7.4.3 The routing is the part that worked

The run after those four changes reached **46.8 % against the same 43.1 % rule, so 3.7 more correct answers per hundred**. It is the first run of the whole project to clear “repeat the last antenna” on a complete test set rather than on a hand-picked segment. The rule itself did not move, which is the point: same people, same protocol, same architecture, and the only difference is which mistakes the training was told to care about and which numbers were allowed to answer.

Figure 7.4 shows where that 3.7 lands, one bar per kind of user. **The gain ranks in exactly the reverse order of the rule, with no exception among the four.** Read against Chapter 5, this is that chapter's finding arriving from the other direction: there, difficulty was shown to live in the person; here, the model is shown to earn its cost precisely where the trivial rule fails, and to earn nothing where it already works.

This number is not the report's headline. It is what the configuration is worth on four hundred training people, and it raises the next question: four hundred people split across four kinds leaves the smallest kind with a few dozen members, which is barely a profile at all. The configuration itself, hour symbols plus one set of numbers per kind of user, is carried unchanged into Chapter 8.

⁴**Cross-training:** training one set of numbers on several people at once, so that people who behave alike inform each other instead of being learnt as independent sequences.

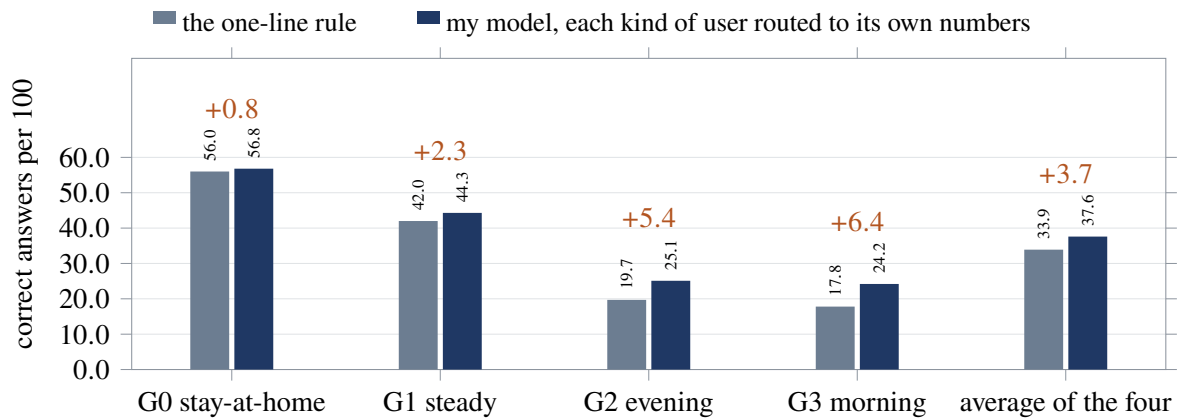


Figure 7.4: The gain, kind of user by kind of user

The result is the ordering, and it is exact. One pair of bars per kind of user, plus the plain average of the four on the right. Grey-blue is the one-line rule, dark blue my model, both on exactly the same moments of the same 100 held-out people, over every position of their day. The gain ranks in the precise reverse order of the rule, with no exception among the four. The stay-at-home user, whom repetition already predicts 56.0 % of the time, gains 0.8; the morning-moving user, whom repetition predicts less than one time in five, gains 6.4; the steady user gains 2.3 and the evening-moving user 5.4.

Two qualifications belong with the headline. The 3.7 is the plain average over four kinds counted once each; weighted by how many people each kind holds it is 3.3, and that is the figure an operator would experience, because 28 % of the population sits in the kind that needed no model at all. And the mechanism depends on the kind of user being identifiable, which Figure 7.2(b) shows collapsing below roughly 60 % of a day of history.

↪ Produced by `train_cellid_llm.ipynb`, the per-group adapter-routing run.

7.5 Conclusion

This chapter answered both questions it opened with. People can be sorted into kinds of user from when they change location, using only the part of the day a predictor will actually have in hand, and the sorting is not decoration: knowing nothing about somebody except which of the four kinds they fall into accounts for 72 % of the spread in how predictable they are, and the stretches of the day derived per kind separate the one-line rule by 27.4 correct answers per hundred where the guessed global stretch separated it by two tenths. And telling the model which kind of user it is looking at does improve the prediction, but only once the kind of user stopped being a symbol in the input and became a switch deciding which trained numbers run. The result is the first run of the project to clear the one-line rule on a complete test set: 46.8 % against 43.1 %, so 3.7 more correct answers per hundred.

This is the chapter I would defend rather than qualify. Cross-training is the largest success of the internship: instead of treating four hundred people as four hundred independent sequences, it lets people who behave alike inform one another, and it is the only change whose effect is clearly larger than the noise around it. What makes it convincing is not the headline but the diagnosis that preceded it. The first attempt failed, and the three-way check, correct symbol, wrong symbol and no symbol at all, landed within two tenths of each other and said exactly where: not in the sorting, not in the hours, but in the channel. Without that check I would have concluded that the sorting was worthless and thrown away the only idea in the

report that worked.

Two things temper the headline rather than undo it. The 3.7 is the plain average over four profiles counted once each, while an operator, who meets the four kinds in their real proportions, would experience 3.3; given the choice again I would lead with 3.3. And four sets of trained numbers are four times the trainable capacity of one, so part of the gain may be capacity rather than the sorting. A single set carrying as many trainable numbers as the four together would separate the two in one run of half an hour, and that control was never run.

8 The impact of the number of users

8.1 The goal

Four hundred training people split across four kinds of user leaves the smallest kind with a few dozen members, and a profile estimated from a few dozen people is barely a profile at all. If the mechanism that produced the project's only real gain depends on those kinds being well populated, then the obvious next lever is the number of users. Hence the question: *how many people does a model need before the population itself is described, and what happens after that point?* The question is unusually well posed on this data, because the unit is one day per person, so the only thing that can be increased is the number of *people*, never the amount of history per person.

Where the numbers in this chapter come from. Three sets of people of Table 3.1. The 400-person point is the `MOBILE-USER DAY`, graded on 100 held-out people against a 43.1 % rule. The 900 to 1,900-person points are the `2,000-USER DAY`, graded on 100 held-out people against a 48.2 % rule. And the 5,000 to 21,000-person points are the `ELEVEN-DAY SET`, graded on **1,000 held-out people** against a 50.2 % rule. In every case the test people are people the model has never trained on, and because those three rules differ, this chapter never compares two scores directly. It compares distances.

8.2 Eight training sizes, everything else fixed

The proper version of this experiment fixes everything except the number of people. Same architecture, same settings, same cut of the day, hour symbols and the four kinds of user active throughout, and each size read against the one-line rule measured on its own test people.

The largest sizes required data built for the purpose: eleven day files of 2,000 people each, stacked into 22,000 person-days, of which 1,000 are held out. The sizes are nested, so a larger one is strictly *more* data and never *other* data; without that, a difference between two sizes could have come from *which* people were drawn rather than from how many. Over those sizes the number of records grows from 369,451 to 1,538,379 and the number of available training slices¹ from 5,379 to 22,522, for a total of two hours fifty-one minutes of computing time.

Figure 8.1 puts the eight sizes on one axis, each read against the one-line rule measured on its own test people, and the curve it draws has two halves: it climbs steeply up to about two thousand people, and then, over a further ten-fold increase, it does not move again.

8.2.1 The flat part is real, not noise

“The best size is followed by the worst” is the signature of noise, but a signature is not a test. The test used is a *paired resampling* of the 1,000 held-out people.

Table 8.1 reports it, and as its last column shows, every interval contains zero: each individual step, and the interval spanning the entire range from 5,000 to 21,000 people, whose best estimate is 0.04. So the correct statement is not “the gain is small” but “**no gain was**

¹**Training slice:** one continuous stretch of a person's day, used as a single training example.

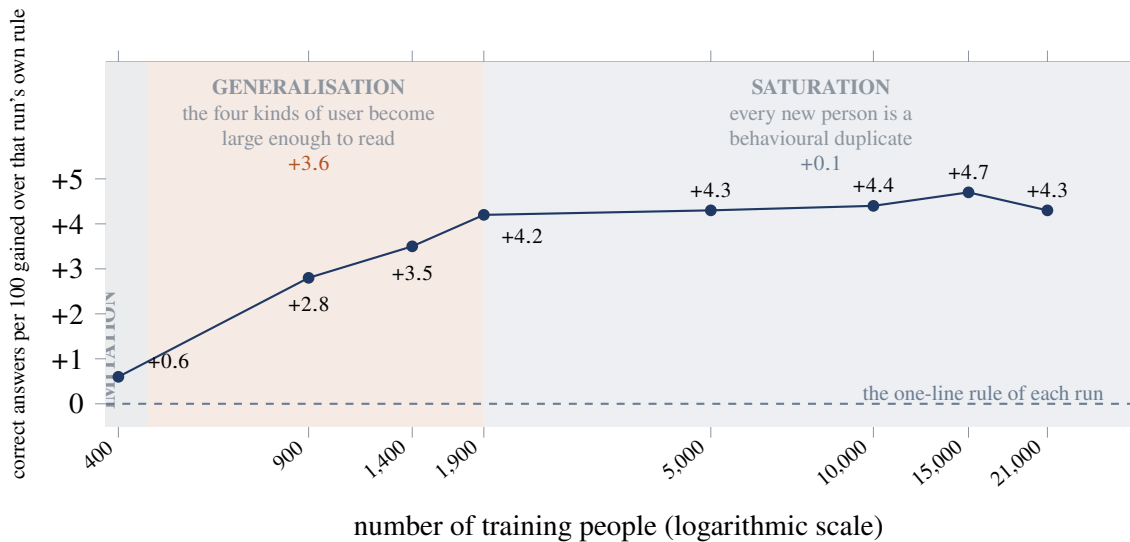


Figure 8.1: Eight training sizes: the gain climbs, then stops

What is on each axis. Eight training runs differing only in how many people they were trained on. The horizontal axis is that number on a logarithmic scale, so equal distances are equal multiples. The vertical axis is *not* a score: it is the distance between the model and the one-line rule on that run’s own test people, the only quantity that survives a change of test set. The dashed line at zero means “indistinguishable from repeating the last antenna”.

The middle band holds everything, and two things constrain how far it reads. At 400 people the model gains 0.6, which is to say nothing. Between 400 and 1,900 the gain climbs 0.6, 2.8, 3.5, 4.2, essentially the entire benefit this project ever obtained from data volume. The ten-fold increase that follows buys 0.1, and the best size, 15,000, is followed by the worst, 21,000: the curve does not slow down, it stops. Two caveats. The points do not come from the same test people, which is why the axis is a distance and the small step across that boundary is not a measurement. And every size received the same budget of training time, so at 21,000 people only 36 % of the material was ever read: the defensible claim is that 1,900 people are enough *at this budget*.

↪ Produced by `train_cellid_final.ipynb`, the eight-tier scaling notebook.

measured”. Nor does anything hide behind the overall average. Table 8.2 takes the same sizes through six different measurements, and the spread across the whole range is between 0.2 and 1.3 on every single line. Table 8.3 closes the last way out: the flat half of the curve stays flat at every cut of the day, so the gain does not come from handing the model a longer history either.

8.2.2 The one thing that does move the score

One thing does, and it is a property of the person rather than of the dataset: which kind of user they are. Evaluated with the 21,000-person model on the same held-out people, the four kinds score 62.8 % for the stay-at-home user, who is 22 % of the people in ELEVEN-DAY SET; 53.6 % for the steady user, 37 %; 43.0 % for the evening-moving user, 22 %; and 40.4 % for the morning-moving user, 18 %. That is a spread of **22.4 correct answers per hundred**. The shares are not those measured on MOBILE-USER DAY in Chapter 7, since these are twenty-one thousand different people, but the four shapes themselves were essentially identical at every

size, which is itself a result about scale.

Put side by side: **22.4 against 0.4**. Who the person is outweighs how much data was used to train, by a factor of more than fifty.

Table 8.1: No step up the ladder is real: every interval contains zero

Step in training people	Measured difference	Range it could really be	Real gain?
5,000 → 10,000	+0.20 %	-0.21 ... + 0.64	no
10,000 → 15,000	+0.09 %	-0.32 ... + 0.52	no
15,000 → 21,000	-0.25 %	-0.72 ... + 0.20	no
5,000 → 21,000 (<i>total</i>)	+0.04 %	-0.43 ... + 0.51	no

The question, and how it is tested. Between two training sizes the score moves a little. Is that the model genuinely improving, or the luck of which thousand people happened to be in the test set? A thousand people are drawn at random from the thousand held-out ones, with replacement; both models are scored on exactly that draw and the difference noted; that is repeated two thousand times, and the middle 95 % of the differences is the range in the third column. Because both models always see the same draw, the luck of the sample cancels.

What that licenses me to say. If a range contains zero, “no difference at all” is a perfectly ordinary outcome and the two sizes cannot be told apart. Every row contains zero, including the last, which spans the whole ladder from 5,000 to 21,000 people and whose best estimate is 0.04. So the claim is not “the gain is small” but *no gain was measured*. The fitted scaling law agrees: a slope of +0.18 % per ten-fold increase in users, on a line accounting for only 12 % of the spread of the eight points.

↪ Produced by `train_cellid_final.ipynb`, the paired-bootstrap cells, 2,000 resamples.

Table 8.2: Six ways of measuring, and all six say the same thing

Way of measuring	1,900 [†]	5,000	10,000	15,000	21,000	Spread
Top-1, the headline	52.4 %	54.5 %	54.6 %	54.9 %	54.5 %	0.4 %
Top-3	71.4 %	72.0 %	72.1 %	72.1 %	72.2 %	0.2 %
Top-5	78.6 %	78.9 %	78.9 %	78.9 %	79.3 %	0.4 %
Top-1 on a cold start, 30 % of the day	51.4 %	52.3 %	52.6 %	52.3 %	52.1 %	0.5 %
The middle person		49.1 %	49.2 %	49.2 %	50.3 %	1.2 %
The worst tenth of people		14.4 %	13.1 %	14.4 %	13.1 %	1.3 %

Why six rows and not one. A flat headline can hide movement underneath. **Top-1** is the strict headline; **top-3** and **top-5** ask whether the right antenna is anywhere in the first three or five. **Cold start** gives the model only 30 % of each person’s day. **The middle person** ignores the busiest people entirely, and **the worst tenth** guards against a model that lifts its average by abandoning the people it already served badly.

The verdict is the last column. Across four training sizes covering a four-fold increase in people, no row moves by more than 1.3 correct answers per hundred. One row is worth isolating: on a cold start the model beats its rule by 4.6 against 4.3 at 80 % of the day, so its edge is *largest* on people about whom little is known, which is the useful case, and that edge too is identical at 5,000 and at 21,000 people. The 1,900 column, marked [†], is measured on 2,000-USER DAY against a 48.2 % rule: a reference point, not a fifth size, and the **Spread** column covers the four ELEVEN-DAY SET sizes only.

↪ Produced by `train_cellid_final.ipynb`, the six-metric evaluation of the four tiers.

Table 8.3: The gain does not come from giving the model a long history

Past given	Moments	The rule	900 people	1,900 people	Difference
30 %	5,491	47.1 %	50.0 % (+2.9)	51.4 % (+4.3)	+1.4
40 %	4,703	46.8 %	49.9 % (+3.1)	51.4 % (+4.6)	+1.5
50 %	3,922	46.6 %	50.1 % (+3.5)	51.5 % (+4.9)	+1.4
60 %	3,142	47.0 %	50.3 % (+3.3)	51.2 % (+4.2)	+0.9
70 %	2,359	47.3 %	50.5 % (+3.2)	51.6 % (+4.3)	+1.1
80 % (<i>ref.</i>)	1,570	48.2 %	51.0 % (+2.8)	52.4 % (+4.2)	+1.4
90 %	781	50.3 %	51.9 % (+1.6)	53.6 % (+3.3)	+1.7

What is being varied, and how to read a row. Two trained models, one on 900 training people and one on 1,900, both graded on the same 100 held-out people, and each row hands them a different amount of that person’s day as history. **Moments** is how many graded moments the row rests on, **the rule** is what “repeat the last antenna” scores on them, the two model columns give the score with the distance to the rule in brackets, and **Difference** is how much the larger training set was worth on that row.

Two independent levers. The 1,900-person model is ahead of the rule by 4.3 having seen only 30 % of the day, and by 4.2 having seen 80 %, so what it contributes is essentially independent of how much history it is handed. And going from 900 to 1,900 training people is worth between 0.9 and 1.7 on every row alike. 2,000-USER DAY, same 100 held-out people for both sizes.

↪ Produced by `train_cellid_final.ipynb`, the context-fraction evaluation at the smaller tiers.

8.3 The reason the curve climbs, then stops

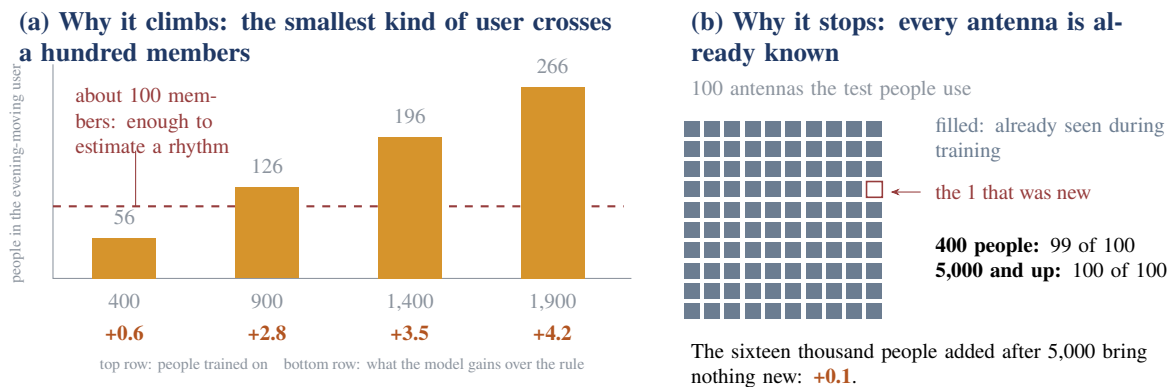


Figure 8.2: The reason the curve climbs, and the reason it stops

Panel (a): one mechanism fits the timing of the jump. The bars are how many people fall in the least populated of the four kinds, the evening-moving user, at 14 % of the population. Under each bar is what the model gained over the one-line rule at that training size. The count crosses roughly a hundred members exactly between the two sizes where the gain jumps from 0.6 to 2.8. A label can only carry information if the group it names has enough inhabitants for its rhythm to be estimated at all, which is precisely what the failed run of Chapter 7 showed.

Panel (b): above the threshold every new person is a duplicate. Each square is one of a hundred antennas the test people actually use. At the smallest training size the model has already met 99 of them, and from 5,000 people onward, all of them; the four rhythms of Figure 7.1 keep the same shape at every size as well. So more people can only bring more of the same. This is an explanation that fits, not a proof: the alternative reading, that the training budget rather than the population ran out, is stated in Section 8.3.1.

↪ Produced by `train_cellid_final.ipynb`, the eight-tier scaling notebook; the member counts are the group shares of Chapter 7 applied to each size.

Figure 8.2 gives the two halves of the answer: its panel (a), that the smallest of the four kinds of user only becomes populated enough to describe at the size where the score jumps, and its panel (b), that by then the model has already met every antenna the test people use, so a further person can only bring more of the same. Below the threshold, copying is simply the cheapest way to score well. On the 7,745 graded moments of the 100-person test set, 48 out of every hundred ask for the same antenna again, so half the task falls without modelling anything. Meanwhile 400 training people split across four kinds of user and four stretches of the day leaves roughly twenty-five examples per cell of that grid, and twenty-five examples cannot tell a habit from a coincidence. A gain of 0.6 is the signature of a model that has learned the format and the trivial habit and nothing about behaviour, which is exactly what the checks of Chapter 7 independently concluded.

8.3.1 Two limits of this experiment

The flat half of the curve was measured at *constant computing time*, and that is a deliberate choice with a consequence. Every size received the same budget: enough steps to read 8,000 training slices, however many were available. At 5,000 people there are 5,379, so the run gets

through all of them one and a half times. At 21,000 there are 22,522, so 8,000 are read and **14,522 are never read even once**, which is 36 % coverage. The larger dataset was not added; it was spread thinner. Separately, only about 2 numbers in every 100 were ever allowed to change, and that fraction was never increased with the data either.

So the defensible conclusion is “*with this much training time and this much trainable capacity, 1,900 people are enough*”, and not “more people can never help”. Both statements would draw the same flat line, and one experiment would tell them apart: give the largest dataset the training time its size deserves. That run was not performed, and the claim is limited accordingly.

8.4 The population is cheap, the person is not

Data volume is a real lever over one narrow stretch and worthless above it. From roughly 400 to 1,900 training people the gain climbs from 0.6 to 4.2, and above 1,900 it does not move at all: 0.1 across a ten-fold increase, with zero inside every confidence interval and six different measurements agreeing. For anyone who would pick this work up, two thousand people is the operating point.

And what difficulty there is lives in the person. 22.4 correct answers per hundred separate the four kinds of user, against 0.4 across the whole range of training sizes. The ratio between those two numbers is more than fifty, and it is the most robust finding in this report.

Read together, the two say where the remaining work is. It is not in collecting more people, because that is exhausted at two thousand. It is in conditioning better on who each person is, and specifically in doing so for somebody about whom little is yet known.

8.5 A side experiment: levelling records instead of adding users

One alternative to adding people was tested in the same iteration. The idea, adapted from a colleague’s work, is that a few very long days dominate training: with natural day lengths, the longest quarter of people carries 44.3 % of the training positions,² so a fine-tune spends most of its budget learning the habits of the heaviest users. The remedy is to stretch every day to exactly 200 records by repeating each existing record a few times, spread evenly along the day.

It delivers exactly what it promises. On the 400 training people the ratio between the longest and the shortest day goes from 200:19 to 1:1 and the inequality of per-person weight³ from 0.294 to 0.

Figure 8.3 is where the technique is judged, and panel (a) is the warning: the levelled run posts by a wide margin the highest absolute score of the whole project and is nonetheless behind its own one-line rule. Panel (b) says why. The redistribution is paid for exactly as one would fear: the shortest quarter of people comes out level, 1.1 behind on days that are

²**Training position:** one graded antenna name in the training data. Somebody with 200 events contributes about 200 of them and somebody with 20 contributes 20, so long days dominate by being long.

³**Gini coefficient:** a measure of inequality borrowed from economics, running from 0 when everybody contributes the same number of training positions to 1 when a single person contributes all of them.

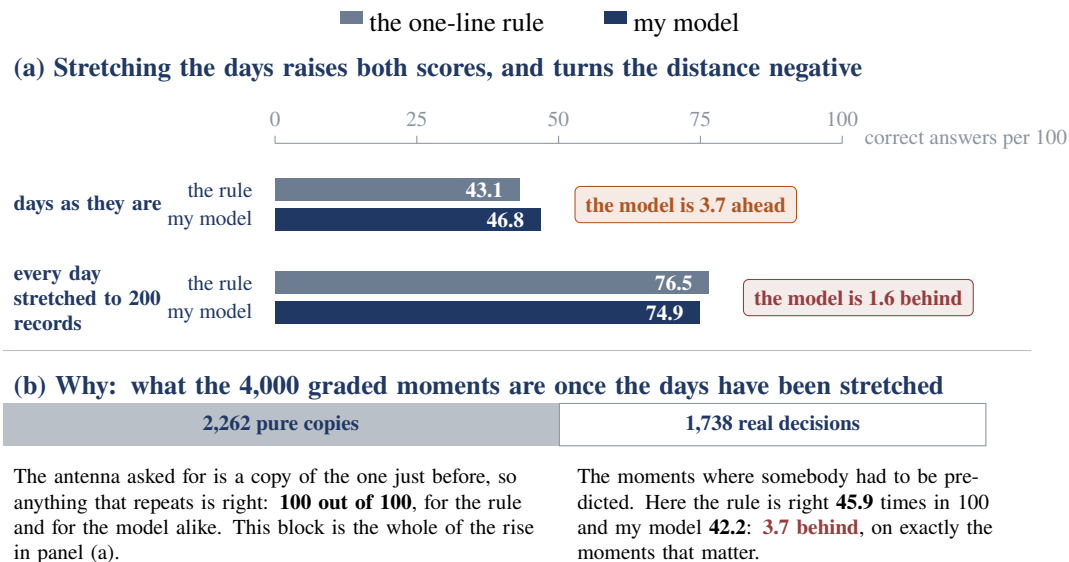


Figure 8.3: Stretching every day to 200 records: a high score, a worse result

Panel (a): a spectacular score, and a result that changed sign. Every person's day is stretched to exactly 200 records by repeating records already there, so that the longest quarter of people stops carrying 44 % of the training material. Each run is shown against *its own* one-line rule, and only the comparison inside a pair means anything. The levelled model reaches 74.9 %, by a wide margin the highest absolute score of the project, but the rule on those same stretched days reaches 76.5 %.

Panel (b): the reason is arithmetic, not modelling. Stretching a day fills it with copies, and a copy is a moment on which repeating the last antenna cannot be wrong. That is why both scores rose by about thirty. On the 1,738 moments that are real decisions the model is behind. Its bar there is derived from the reported 74.9 % assuming it never misses a copy, and it stays below the rule at either end of that assumption. And because a copied step never changes location, levelling compresses the spread between people by a factor of 2.3, destroying the very signal the four kinds of user are cut on.

82 % copies, while the longest quarter loses 3.6 on days that are 22 % copies. Weight was transferred from the people who had the most to say to the people who had the least. The technique is not wrong and the imbalance it fixes is genuine, but on this data it removes more information than it redistributes.

8.6 Conclusion

The chapter asked how many people a model of this kind needs, and it has an answer with a number in it: about two thousand. Below that, the number of people is a real lever and the edge over the one-line rule grows from 0.6 to 4.2 more correct answers per hundred. Above it, a ten-fold increase to twenty-one thousand people buys 0.1, with zero inside every confidence interval, on six different measurements and at every cut of the day. That claim is bounded in one way that matters and the chapter states it plainly: every size was given the same computing time, so at 21,000 people only 36 % of the available material was ever read. The honest form of the conclusion is “at this training budget, and with this small a part of the model allowed to change, 1,900 people are enough”, not “more people can never help”. One run would tell

the two apart and it was not performed.

Beside that sits the number I would keep if I had to keep one from the whole report. The four kinds of user are 22.4 correct answers per hundred apart on the very same trained model, against 0.4 across the entire range of training sizes. Who the person is outweighs how much data was used to train by a factor of more than fifty, and the side experiment on levelling record counts points the same way: making everybody weigh the same in training moved weight from the people who had the most to say to the people who had the least.

Finding the operating point is the result I am most satisfied with, because it is a statement about the problem rather than about my own pipeline, and the eight sizes together cost under three hours of computing time. What I would change is *when* it was measured. Most of the internship was spent improving the model on four hundred training people, a size this very curve later placed inside the region where a model can do little but imitate the trivial rule, and measuring the curve in week two would have moved the whole project onto two thousand people from the start. There is one reason the order was not simply wrong: the explanation for where the curve stops climbing only exists because the four kinds of user had already been built and measured.

9 Conclusion

The question given to me at the start of this internship was whether a language model, retrained on anonymous network records, can predict which antenna a person's phone will be attached to next. Nine weeks and eleven iterations later the answer is yes, by a modest and measurable margin, and the more interesting part of the answer is the conditions under which that is true.

9.1 The three findings the work settles

Three findings came out of the nine weeks, and Figure 9.1 states each of them as a single number: the edge over the one-line rule, the number of people past which collecting more stops paying, and the distance between the easiest and the hardest kind of user.

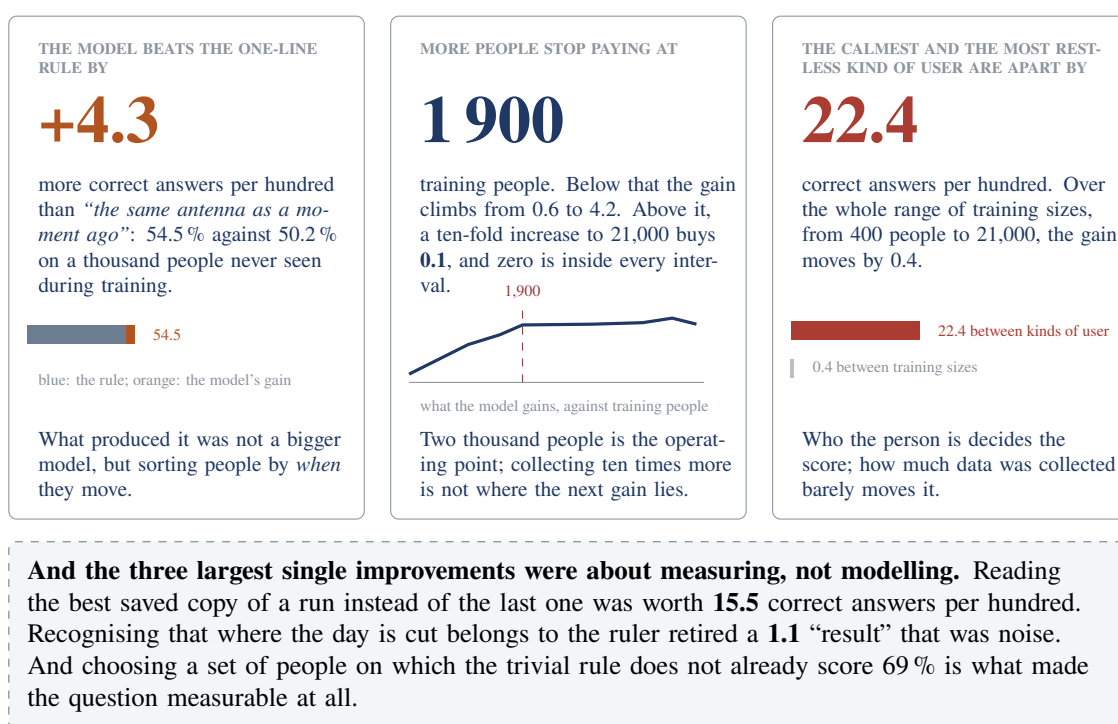


Figure 9.1: The whole report in three numbers

Why these three and not others. Each is the answer to one of the three questions the internship actually asked: does a retrained language model beat the trivial rule, how much data does that take, and where does the remaining difficulty live. They are ordered by how firmly they are established. The **+4.3** is one measurement on one set of people, repeated once on a second group; the **1,900** rests on eight training sizes and a resampling test; the **22.4** is the most robust of the three, because it appears in every chapter that split its results by person.

What the dashed band is doing there. On a task whose honest yardstick is one line of code, three corrections to how things were measured were each worth more than any modelling idea I had. That is not a criticism of the modelling; it is where I would tell somebody picking this up to look first.

↪ Produced by `train_cellid_final.ipynb`, with every number traced to its run in Table B.2.

The model does beat the one-line rule, and it is clear where the gain comes from. A fine-tuned Qwen2.5-0.5B answers correctly **4.3 more times out of a hundred** than “repeat the last antenna”. That number is not large, and it should not be: the competitor is one line of code that is right roughly half the time on people who genuinely move, and for five of the eleven iterations nothing I trained beat it at all. What made the difference was neither capacity nor volume but *conditioning on who the person is*, sorting people by the hours at which they actually change location and sending each kind of user to its own set of trained numbers. That single change took the model from 1.8 behind the rule to 3.7 ahead of it, and the gain fell in the exact inverse order of how well repetition already worked for each of the four kinds.

The amount of data needed is small, and the report says where it stops paying. Population size is a real lever from 400 to 1,900 training people, where the gain grows from 0.6 to 4.2, and is worth nothing above it. One explanation fits the timing: the least populated of the four kinds of user crosses roughly a hundred members exactly between the two sizes where the score moves.

Difficulty lives in the person, not in the dataset. This is the finding I would keep if I had to keep one. The score of an individual person ranges from zero to one hundred per cent, and what predicts it is how repetitive that person’s movement already is. An average score on this problem is a weighted mixture of segments of wildly different difficulty, and quoting it alone hides the only part of the result that is about the model.

9.2 The work left undone for lack of time

The first three items are unfinished work rather than future research. Each could be done in under a day, and I would do them before anything else.

1. **Re-score the earlier model on the later test people.** One grading pass, no training. Because the set of test people changed four times, the 3.7 of iteration 8 and the 4.2 of iteration 9 are two separate measurements of the same effect rather than a progression. Re-scoring puts every row of the project on one axis.
2. **Train the same configuration twice from different random starting points.** About half an hour. The resampling test of Chapter 8 covers the luck of which test people are drawn, but not the fact that training itself is not perfectly repeatable, so differences of a few tenths between sizes sit on a floor nobody measured.
3. **Run the capacity control for the routing.** Four sets of trained numbers are four times the trainable capacity of one, so part of the project’s main gain may be capacity rather than the sorting itself. A single set carrying as many trainable numbers as the four together, on the same data, separates the two.

Two experiments were prepared and did not complete. A seven-day comparison, one independent training per weekday, was launched and the session crashed after five hours with none of the seven finished, so this report claims no effect of the day of the week. And the scaling curve was measured at constant computing time, which means that at 21,000 people only 36 % of the available training material was ever read; giving the largest dataset the

training time its size deserves is the one experiment that would separate “1,900 people are enough at this budget” from “more people never help”.

9.3 Three directions worth taking next

Make the conditioning work on a short history. This is the limitation I would put first if someone asked what the work is good for. The model’s edge is largest on people about whom little is known, 4.6 with only 30 % of a day in hand against 4.3 with 80 %, and that could start is exactly the case an operator cares about. But the kind of user is identified correctly for 92 out of 100 people at the reference protocol and only 61 and 44 out of 100 when half a day or a third of it is available. So the very conditioning that produces the edge decays as the history shortens. Building an identifier that works on a short prefix is a smaller and better-posed problem than the one I spent the internship on, and it is where I would start again.

Widen the question from one day to several. The question given to me was a within-day one, and that is the question this report answers. The obvious continuation is to predict day $N+1$ from days 1 to N , on data whose identifiers are stable from one day to the next: observing the same individual’s home-to-activity move several times, and anything that could be called personalisation. That is the only item on this list that changes what the problem *is* rather than how well it is measured.

And check whether any of this survives a different city. Every result here comes from a single Czech region over fifteen days in March 2014, with 369 antennas over 113 masts. A denser urban area with thousands of antennas could behave differently, and the saturation result in particular partly reflects how small this world is.

9.4 In one paragraph

On single-day cellular records, what a population has in common is cheap to learn, it is measurable, it is worth about four correct answers per hundred over a trivial rule, and it is exhausted at around two thousand people. What it is not yet is usable on somebody who has just appeared, since the conditioning it rests on has to be read from most of a day of history. Closing that gap, rather than training a larger model on a larger corpus, is where I would start again.

A Timeline of the eleven iterations

The body is organised by question rather than by date. This appendix restores the chronology. Figure A.1 gives one entry per iteration, with the question it asked, what changed, and the number it produced.

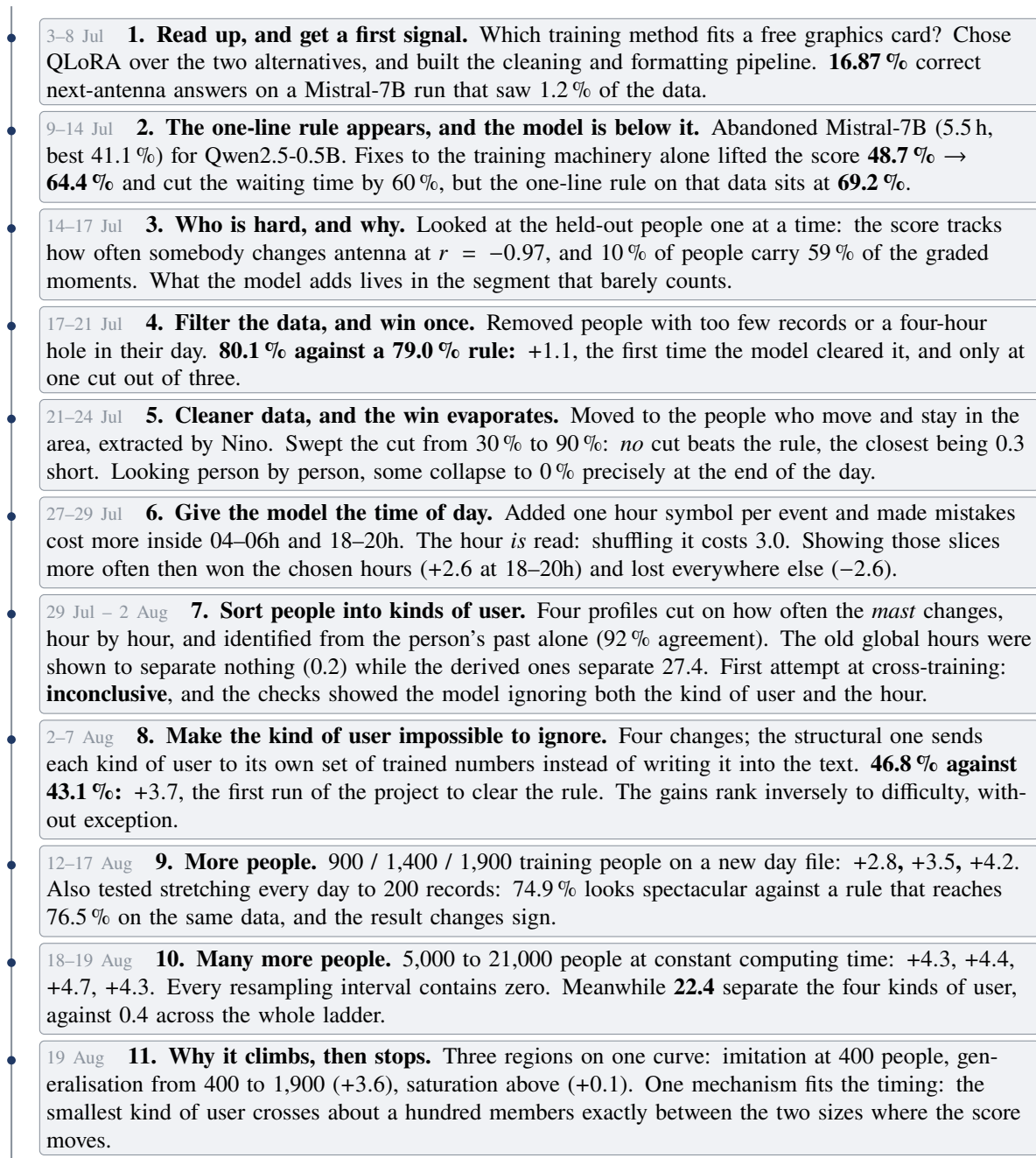


Figure A.1: Timeline of the eleven iterations

What this is, and how to read the numbers. The eleven iterations in order, with the question each asked and the number it produced. Read the *distances* to the one-line rule, never the absolute scores: the set of people changed four times over the period, and with it the difficulty of the test group, so 64.4 % in iteration 2 and 46.8 % in iteration 8 are not comparable at all, whereas “4.8 behind the rule” and “3.7 ahead of it” are. Five iterations sit below the trivial rule, one accidental win at iteration 4 was removed by the next, and a real one arrives from iteration 8 onward.

↔ Produced by `train_cellid_llm.ipynb`, iterations 1–8; iterations 9–11 come from `train_cellid_final.ipynb` and `train_cellid_week.ipynb`, and every number is traced in Table B.2.

B Reproducibility

Three tables record what would be needed to reproduce the results: the training settings of every family of runs, the run behind every number quoted in the body, and the configuration keys that encode a decision argued somewhere in the report rather than a preference. Table B.1 gives the settings, Table B.2 the origin of every number quoted in the body, and Table B.3 the eleven keys that encode an argued decision. The sets of people themselves are described in Table 3.1.

Table B.1: The training settings of each family of runs

	Iter. 1	Iter. 2–6	Iter. 7–9	Iter. 10–11
Base model	Mistral-7B-v0.1	Qwen2.5-0.5B-Instruct	Qwen2.5-0.5B-Instruct	Qwen2.5-0.5B-Instruct
Loading	4-bit (QLoRA)	bf16 / fp32	bf16 / fp32	bf16 / fp32
LoRA rank r	16	16	16	16
LoRA α	32	32	32	32
LoRA dropout		0.1	0.1	0.1
Learning rate	default	1.5×10^{-4}	1.5×10^{-4}	1.5×10^{-4}
Weight decay		0.01	0.01	0.01
Max passes	100 steps	20	20	20 (cosine horizon)
Early stopping		patience 8, then 4	patience 4	4 evaluations
Eval. sequence length		512 tokens	512 tokens	512 tokens
Training window		128 events, stride 64	128 events, stride 64	128 events, stride 64
Context fraction		0.8	0.8	0.8
Added symbols	antenna names	369 antennas	369 antennas + 24 hours	369 antennas + 24 hours
Transition sampler		$\times 3$ from iter. 6	$\times 3$	$\times 3$
Emphasised hours		global (4–6, 18–20)	per kind of user	per kind of user
Kinds of user			4, read from the past	4, refitted at each size
Stopping protocol	fixed steps	passes	passes	1,000 optimiser steps
Seed		42	42	42
Hardware	Colab T4	Colab T4	Colab T4	Colab T4, 42 min/size

How to read it. One column per family of runs, in chronological order. Everything not listed as changing was held constant on purpose, so a difference between two runs is a difference in the data or in the training signal, never in the recipe. A blank cell means the setting did not exist in that family.

The three rows that matter most. **Max passes** with **Early stopping** is the rule of Chapter 4: twenty passes are allowed, the run stops as soon as its score on unseen people stops improving, and the best copy is kept. **Context fraction** at 0.8 is where each day is cut, and belongs to the measurement rather than here. **Emphasised hours** is where Chapters 6 and 7 differ, and it is the only substantive change in the last two columns.

Table B.2: The origin of each number quoted in the report

Number quoted in the report	Data	Iter.	Which run produced it
16.87 % first signal	RAW DAY	1	single-notebook Mistral-7B QLoRA run, 100 steps
48.7 % $\rightarrow$ 64.4 %	RAW DAY	2	same notebook, keeping the best saved copy
$r = -0.971$ on the switch rate	RAW DAY, 500 people	2	per-person analysis of the 1,173-person grading
+1.1 at the 80/20 cut	FILTERED DAY	4	training run on the filtered day
No cut beats the rule	MOBILE-USER DAY	5	one trained model, seven grading passes
-3.0 from shuffling the hours	MOBILE-USER DAY	6	re-grading the same trained copy
+2.6 at 18–20h	MOBILE-USER DAY	6	the targeted-sampling run, whole day graded
27.4 separation of the derived hours	MOBILE-USER DAY	7	analysis of the kinds of user on the training file
+3.7, first run above the rule	MOBILE-USER DAY	8	routing each kind of user to its own numbers
+4.2 at 1,900 training people	2,000-USER DAY	9	scaling run on the 2,000-person day
-1.6 on stretched days	LEVELLED CORPUS	9	the 200-record levelling run
+4.3, flat from 5,000 to 21,000	ELEVEN-DAY SET	10	four-size scaling notebook, constant training time
The resampling intervals	ELEVEN-DAY SET	10	2,000 paired resamples over the test people
0 of 7 weekday trainings finished	ELEVEN-DAY SET	11	seven-day notebook; the session crashed after 5 h

What this table is for. Every number quoted anywhere in the body can be traced back to one run through this table. The **Data** column names the set of people it was measured on, described in Table 3.1, and **Iter.** is the numbered iteration of Appendix A.

Why the data column matters as much as the number. Two rows measured on different sets of people are not comparable on their absolute scores, only on their distances to the rule of that row’s own data. Reading this table alongside Table 3.1 is what makes that check possible.

Table B.3: The settings that encode a decision, not a preference

Key	Value	The decision it encodes
context_fraction	0.8	where each day is cut; part of the ruler (Ch. 4)
max_epochs	20	also the decay horizon: lowering it changes the regime
early_stop_patience	4	the best pass arrives early on this task (Ch. 4)
transition_windows	(4,6,×2), (18,20,×4)	the <i>guessed</i> hours of Ch. 6; fallback only
group_windows	None	derive the hours from the data, per kind of user (Ch. 7)
transition_chunk_oversample	3.0	show slices containing a move three times more often
group_bands	4 stretches	coarse enough for each stretch to be estimable
group_shrinkage	8.0	a thin stretch must not dominate a profile
group_detect_fraction	past only	no leakage: the kind of user is read from the past
max_steps (scaling)	1,000	the same training time at every size (Ch. 8)
val_subsample (scaling)	250 fixed people	the same stopping rule at every size

Why these eleven and not the whole configuration file. None of these is a tuning knob. Each one encodes a decision argued somewhere in the body, so that a reader who changes one knows which claim of the report they are changing. The ordinary settings, learning rate and the rest, are in Table B.1.

C Glossary

The terms that recur from chapter to chapter, each in one or two sentences.

Antenna, mast, cell_id. A *mast* is one physical installation. It carries three or four *antennas* pointing in different directions, and each of those is what the working files call a *cell_id*, for instance UKVCHE102, whose middle letters identify the mast. Mobility is always measured on the mast, because 12.4 out of every 100 back-to-back records change antenna without changing place; what has to be predicted stays the full antenna name.

Cell-switch rate. Out of every hundred back-to-back records of one person, how many land on a different antenna than the one before. Near 0 % for somebody who stays home all day, near 100 % for somebody constantly moving. It is the single property that best predicts how well that person is scored.

Correct answers per hundred. How every result here is expressed. The model gaining 4.3 means it answers correctly 54.5 times out of a hundred where the one-line rule answers correctly 50.2 times, on the same people and moments.

Held-out people. People set aside before training and never shown to the model, on which every reported score is measured.

Hour symbol. A symbol <H00> to <H23> placed before each event. Always handed to the model, never predicted by it, because the current time is known when a prediction is asked for.

Kind of user, or behavioural profile. One of four groups of people, sorted automatically by how often their *mast* changes over four broad stretches of the day, named by increasing mobility: G0 the stay-at-home user, G1 the steady user, G2 the evening-moving user, G3 the morning-moving user.

LoRA, QLoRA. Training methods that freeze the model's original numbers and train a small set of new ones beside them, about 2 in every 100 here. QLoRA additionally stores the frozen numbers in a coarser form.

One-line rule, or persistence baseline. The rule “the next antenna is the last one seen”. No model, no training, and not a constant: 69.2 % on a raw day and 43.1 % on people who genuinely move, because the people differ.

Paired resampling. A test in which the same test people are redrawn at random for both models being compared, so the luck of the draw cancels out. A range containing zero means the two cannot be told apart.

Protocol: last fifth, or whole day. “Last 20 %” grades only the final fifth of a test person's day; “whole day” grades every position of it, which is harder and lowers the one-line rule on the very same people from 45.5 % to 43.1 %.

Record, or event. One antenna name paired with one time. Not a movement: an untouched phone still produces records, because the network asks it to announce itself on a fixed cycle.

Share of the day given, or context fraction. How much of a person's day is handed to the model before it has to produce the rest, 80 % by default. It belongs to the measurement rather than to the model: moving it moves the one-line rule too. Not the model's *context window*, the hard limit of 512 symbols it may look at in one go.

Top-1, top-3, top-5. Whether the antenna the person really visited is the model's first guess, or anywhere among its first three or first five out of the 369 it knows.

Bibliography

- [1] Xu Chen, François Mériaux, and Stefan Valentin. Predicting a user’s next cell with supervised learning based on channel states. In *IEEE 14th International Workshop on Signal Processing Advances in Wireless Communications (SPAWC)*, 2013.
- [2] Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. QLoRA: Efficient finetuning of quantized LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [3] Bradley Efron and Robert J. Tibshirani. *An Introduction to the Bootstrap*. Chapman & Hall, 1993.
- [4] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022.
- [5] Ervin L. Ikanovic and Anders Mollgaard. An alternative approach to the limits of predictability in human mobility. *EPJ Data Science*, 6:12, 2017.
- [6] Jaeseong Jeong, Mathieu Leconte, and Alexandre Proutière. Cluster-aided mobility predictions. In *IEEE INFOCOM 2016: The 35th Annual IEEE International Conference on Computer Communications*, 2016.
- [7] Albert Q. Jiang, Alexandre Sablayrolles, Arthur Mensch, et al. Mistral 7b, 2023. arXiv:2310.06825.
- [8] Massimiliano Luca, Gianni Barlacchi, Bruno Lepri, and Luca Pappalardo. A survey on deep learning for human mobility. *ACM Computing Surveys*, 55(1):1–44, 2021.
- [9] Luca Pappalardo, Filippo Simini, Salvatore Rinzivillo, Dino Pedreschi, Fosca Giannotti, and Albert-László Barabási. Returners and explorers dichotomy in human mobility. *Nature Communications*, 6:8166, 2015.
- [10] Karl Pearson. Note on regression and inheritance in the case of two parents. *Proceedings of the Royal Society of London*, 58:240–242, 1895.
- [11] Zhenlin Qin, Leizhen Wang, Yancheng Ling, Francisco Camara Pereira, and Zhenliang Ma. A foundational individual mobility prediction model based on open-source large language models, 2025. arXiv:2503.16553.
- [12] Qwen Team. Qwen2.5 technical report. Technical report, Alibaba Group, 2025. arXiv:2412.15115.
- [13] Chaoming Song, Zehui Qu, Nicholas Blumm, and Albert-László Barabási. Limits of predictability in human mobility. *Science*, 327(5968):1018–1021, 2010.
- [14] Xinglei Wang, Meng Fang, Zichao Zeng, and Tao Cheng. Where would I go next? large language models as human mobility predictors, 2023. arXiv:2308.15197.